QUESTION ANALYSIS

# EEE459

Compound Semiconductor and Hetero-Junction Devices

# Question Analysis

**Sessions 2021–22 to 2017–18** *4 papers, 82 questions, 148 formulas*

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# 1. At a glance:

This analysis covers **4 papers** (2021-22, 2020-21, 2018-19, 2017-18), with 82 questions and 510 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

### The five topics with the most marks:

| Topic | Marks | Share | Questions | Top subtopics (marks) |
|---|---|---|---|---|
| Bipolar transistors | 108 | 21 % | 16 | HBT (24); Ebers-Moll (22); Injection efficiency (12) |
| Metal-semiconductor contacts | 94 | 18 % | 16 | Schottky junction (53); Ohmic contacts (22); Barrier lowering (14) |
| Heterojunctions | 81 | 16 % | 13 | Band diagrams (30); Fields and potentials (24); HEMT (21) |
| Alloys and strain | 66 | 13 % | 9 | Lattice matching (26); Quaternary alloys (16); GaAsP band gaps (12) |
| JFETs | 63 | 12 % | 10 | Pinch-off (31); Channel profiles (14); Transconductance (14) |

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

# 2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

| Topic and subtopic | 2021-22 | 2020-21 | 2018-19 | 2017-18 | Term tests | All |
|---|---|---|---|---|---|---|
| **Band structure and materials** | · | **17** | **6** | **7** | **12** | **42** |
| Density of states | · | **6** | · | · | **8** | 14 |
| Hole bands | · | **6** | · | · | **4** | 10 |
| Crystal structure | · | · | **6** | **3** | · | 9 |
| Materials | · | **5** | · | · | · | 5 |
| Effective mass | · | · | · | **4** | · | 4 |
| **High-field transport** | · | **11** | · | · | **18** | **29** |
| Mobility | · | **6** | · | · | **15** | 21 |
| Gunn effect | · | **5** | · | · | **3** | 8 |
| **Alloys and strain** | · | **20** | **8** | **27** | **11** | **66** |
| Lattice matching | · | **9** | · | **9** | **8** | 26 |
| Quaternary alloys | · | **7** | · | **9** | · | 16 |
| GaAsP band gaps | · | · | · | **9** | **3** | 12 |
| Epitaxy | · | · | **8** | · | · | 8 |
| Isoelectronic traps | · | **4** | · | · | · | 4 |
| **PN junction** | · | **5** | · | · | · | **5** |
| Graded junction | · | **5** | · | · | · | 5 |
| **Metal-semiconductor contacts** | · | **31** | **18** | **26** | **19** | **94** |
| Schottky junction | · | **13** | **6** | **19** | **15** | 53 |
| Ohmic contacts | · | **4** | **7** | **7** | **4** | 22 |
| Barrier lowering | · | **14** | · | · | · | 14 |
| Interface states | · | · | **5** | · | · | 5 |
| **Heterojunctions** | · | **20** | **34** | **17** | **10** | **81** |
| Band diagrams | · | **8** | **5** | **7** | **10** | 30 |


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| Topic and subtopic | 2021-22 | 2020-21 | 2018-19 | 2017-18 | Term tests | All |
|---|---|---|---|---|---|---|
| Fields and potentials | · | **12** | **12** | · | · | 24 |
| HEMT | · | · | **11** | **10** | · | 21 |
| Resonant tunnelling | · | · | **6** | · | · | 6 |
| **MOS capacitor** | · | · | **12** | · | **10** | **22** |
| MOS capacitor | · | · | **6** | · | **5** | 11 |
| Threshold voltage | · | · | **6** | · | **5** | 11 |
| **Bipolar transistors** | · | **17** | **51** | **40** | · | **108** |
| HBT | · | · | **13** | **11** | · | 24 |
| Ebers-Moll | · | **9** | **5** | **8** | · | 22 |
| Injection efficiency | · | · | **12** | · | · | 12 |
| Breakdown | · | · | · | **9** | · | 9 |
| Bandgap narrowing | · | · | · | **8** | · | 8 |
| Base minority carriers | · | · | **8** | · | · | 8 |
| High-level injection | · | **8** | · | · | · | 8 |
| Gummel-Poon | · | · | **7** | · | · | 7 |
| Non-ideal effects | · | · | **6** | · | · | 6 |
| TEBT | · | · | · | **4** | · | 4 |
| **JFETs** | · | **29** | **9** | **25** | · | **63** |
| Pinch-off | · | **15** | **5** | **11** | · | 31 |
| Channel profiles | · | **6** | · | **8** | · | 14 |
| Transconductance | · | **8** | · | **6** | · | 14 |
| Cut-off frequency | · | · | **4** | · | · | 4 |

# 3. Priority list:

**Must know:** Metal-semiconductor contacts (94), Alloys and strain (66), Band structure and materials (42)

**Likely:** Bipolar transistors (108), Heterojunctions (81), JFETs (63), High-field transport (29)

**Less often:** MOS capacitor (22), PN junction (5)

**The rule:** Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.


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# 4. Topic pages:

## 4.1. Band structure and materials:

42 marks, 8 questions, in 4 of 4 papers. Book: Neamen Ch. 3; Sze Ch. 1.

Most often theory (4 of 8). Subtopics: Crystal structure; Effective mass; Hole bands; Materials; Density of states.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Crystal structure (9) | **Atoms per cell** | corner $\frac{1}{8}$, face $\frac{1}{2}$, edge $\frac{1}{4}$, inside 1 | 18-19 1(a) |
| | bcc packing | $4r = \sqrt{3}a$, fraction $\sqrt{3}\pi/8 = 0.68$ | 18-19 1(a) |
| | hcp ideal ratio | $c/a = \sqrt{8/3} = 1.633$ | 18-19 1(a) |
| | Zinc blende bond | $d = \sqrt{3}a/4$ | 18-19 1(a) |
| Effective mass (4) | **Effective mass** | $\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$ | 17-18 1(b) |
| | Band edge parabola | $E - E_c = \frac{\hbar^2 k^2}{2m^*}$ | 17-18 1(b) |
| | Finite difference | $\frac{d^2E}{dk^2} \approx \frac{E_{i+1} - 2E_i + E_{i-1}}{(\Delta k)^2}$ | 17-18 1(b) |
| Hole bands (10) | **Parabolic band** | $E = \frac{\hbar^2 k^2}{2m^*}$ | 20-21 1(c), 20-21 7(c) |
| | Slope and curvature | $\frac{dE}{dk} = \frac{\hbar^2 k}{m^*}$, $\frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$ | 20-21 1(c), 20-21 7(c) |
| | Si hole masses | $m_{hh,lh}[100] = \frac{m_0}{\gamma_1 \mp 2\gamma_2}$, $m_{hh,lh}[111] = \frac{m_0}{\gamma_1 \mp 2\gamma_3}$ | 20-21 1(c), 20-21 7(c) |
| Materials (5) | **Emission wavelength** | $\lambda\ (\mu\text{m}) = \frac{1.24}{E_g\ (\text{eV})}$ | 20-21 2(a) |
| Density of states (14) | **Density of states, 3D** | $g_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$ | 20-21 3(b) |
| | Density of states, 2D | $g_{2D}(E) = \frac{m^*}{\pi\hbar^2}$ per subband | 20-21 3(b) |
| | Density of states, 1D | $g_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$ | 20-21 3(b) |
| | **Parabolic band** | $E = E_c + \frac{\hbar^2 k^2}{2m^*}$ | 21-22 7(b) |
| | Allowed $k$ in a box | $k = \frac{2\pi n}{L}$ | 21-22 7(b) |
| | Infinite-well levels | $E_n = \frac{\hbar^2}{2m^*}\left(\frac{n\pi}{L_z}\right)^2$ | 21-22 7(b) |
| | **3D (bulk)** | $D_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$ | 21-22 7(b) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **2D (well)** | $D_{2D}(E) = \frac{m^*}{\pi\hbar^2}\sum_n H(E-E_n)$ | 21-22 7(b) |
| | **1D (wire)** | $D_{1D}(E) = \frac{1}{\pi\hbar}\sum_n \sqrt{\frac{2m^*}{E-E_n}}$ | 21-22 7(b) |
| | 0D (dot) | $D_{0D}(E) = \sum_i 2\,\delta(E-E_i)$ | 21-22 7(b) |
| | General rule | $D \propto (E-E_n)^{d/2-1}$ | 21-22 7(b) |

## Symbols:

$a$ = lattice constant ↩

$r$ = atom radius ↩

$E_i$ = points read from the E–k diagram ↩

$E$ = energy from the band edge (J) ↩

$k$ = wave number ($m^{-1}$) ↩

$\hbar$ = reduced Planck constant (J · s) ↩

$\lambda$ = wavelength of a photon of energy $E_g$ ↩

$g$ = states per unit energy per unit volume

$E_c$ = conduction-band edge ↩

$m^*$ = effective mass (kg) ↩

$L$ = box size ↩

$n$ = integer ↩

$L_z$ = well width ↩

$H$ = unit step ↩

$d$ = number of free directions ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Crystal structure** | | | | |
| 1 | 18-19 1(a) | 6 | diagram | Draw the following crystal structures. (i) Hexagonal close packed crystal structure (ii) Zinc blende crystal structure. (iii) Body centered cubic crystal structure |
| 2 | 17-18 1(a) | 3 | theory | Define mono crystalline, polycrystalline and amorphous materials. |
| **Effective mass** | | | | |
| 3 | 17-18 1(b) | 4 | derivation | What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram. |
| **Hole bands** | | | | |
| 4 | 20-21 1(c) | 6 | derivation | Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes. |
| 5 | 20-21 TT1 7(c) | 4 | derivation repeats 20-21 1(c) | Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than that of heavy holes. |
| **Materials** | | | | |
| 6 | 20-21 2(a) | 5 | theory | Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it. |
| **Density of states** | | | | |
| 7 | 21-22 TT1 7(b) | 8 | theory repeats 20-21 3(b) | Explain how the density of states changes qualitatively as the dimensionality reduces from 3D → 2D → 1D → 0D. Sketch $D(E)$ vs $E$ for each case. |
| 8 | 20-21 3(b) | 6 | theory | Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \to 2D \to 1D$. Sketch $D(E)$ vs $E$ for each case. |

## 4.2. High-field transport:

29 marks, 5 questions, in 2 of 4 papers. Book: Sze Ch. 1, 10.

Most often theory (5 of 5). Subtopics: Mobility; Gunn effect.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Mobility (21) | **Effective mass** | $\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$ | 17-18 7(b) |
| | Si conductivity mass | $m^*_{cn} = \frac{3}{1/m_l + 2/m_t} = 0.26\,m_0$ | 17-18 7(b) |
| | **Mobility** | $\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$ | 17-18 7(b) |
| | High-field mobility | $\mu(E) = v_d/E$; $v_{sat} \approx 10^7$ cm/s | 17-18 7(b) |
| | **Mobility** | $\mu = \frac{e\,\tau}{m^*}$ | 20-21 1(a), 20-21 7(a) |
| Gunn effect (8) | Drift velocity, low field | $v_d = \mu E$ | 20-21 1(b), 20-21 7(b) |
| | Saturation velocity | $v_d \to v_{sat} \approx 10^7$ cm/s | 20-21 1(b) |
| | Gunn oscillation | $f = \frac{v_d}{L}$ | 20-21 1(b), 20-21 7(b) |

### Symbols:

$\mu$ = mobility ($cm^2/V \cdot s$) ↩

$\tau$ = mean time between collisions (s) ↩

$m^*$ = conductivity effective mass (kg) ↩

$v_d$ = drift velocity (cm/s) ↩

$E$ = electric field (V/cm) ↩

$f$ = transit-time frequency (Hz) ↩

$L$ = length of the GaAs sample (cm) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Mobility** | | | | |
| 1 | 20-21 1(a) | 6 | theory repeats 17-18 7(b) | Between Si and GaAs, which has higher electron mobility and why? |
| 2 | 20-21 TT1 7(a) | 3 | theory repeats 20-21 1(a) | Why is the electron mobility of Si less than GaAs? |
| 3 | 17-18 TT1 7(b) | 12 | theory | Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field. |
| **Gunn effect** | | | | |
| 4 | 20-21 1(b) | 5 | theory | Discuss Velocity saturation and Gunn effect . |
| 5 | 20-21 TT1 7(b) | 3 | theory repeats 20-21 1(b) | What is Gunn effect? |



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## 4.3. Alloys and strain:

66 marks, 9 questions, in 4 of 4 papers. Book: Neamen Ch. 14.

Most often theory (6 of 9). Subtopics: GaAsP band gaps; Quaternary alloys; Lattice matching; Epitaxy; Isoelectronic traps.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| GaAsP band gaps (12) | **GaAsP gaps** | $E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ | 17-18 1(c) |
| | **Direct gap of GaAsP** | $E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2$ | 21-22 7(a) |
| | **Indirect gap of GaAsP** | $E_X(x) = 1.900 + 0.144x + 0.211x^2$ | 21-22 7(a) |
| | Crossover | $E_\Gamma(x_c) = E_X(x_c)$ | 21-22 7(a) |
| | Emission wavelength | $\lambda\ (\mu\text{m}) = \dfrac{1.24}{E_g\ (\text{eV})}$ | 21-22 7(a) |
| Quaternary alloys (16) | Photon energy | $E_g = \dfrac{hc}{\lambda} = \dfrac{1.24}{\lambda(\mu\text{m})}\ \text{eV}$ | 17-18 1(d) |
| | Vegard (quaternary) | $Q(x,y) = xyQ_1 + x(1-y)Q_2 + (1-x)yQ_3 + (1-x)(1-y)Q_4$ | 17-18 1(d) |
| | **Misfit** | $f = \dfrac{a_L - a_s}{a_s}$ | 17-18 1(d), 17-18 3(b), 17-18 7(a) |
| | **Critical thickness** | $h_c \approx \dfrac{a_s}{2f}$ | 17-18 1(d), 17-18 3(b), 17-18 7(a) |
| | **Vegard's law, ternary** | $a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$ | 20-21 2(d) |
| | **Quaternary** $In_xGa_{1-x}As_ySb_{1-y}$ | $P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$ | 20-21 2(d) |
| Lattice matching (26) | **Vegard (ternary)** | $a(x) = x\,a_{AC} + (1-x)\,a_{BC}$ | 17-18 7(a) |
| | Matthews–Blakeslee | $h_c = \dfrac{b(1-\nu\cos^2\alpha)}{2\pi f(1+\nu)\cos\lambda}\left[\ln\dfrac{h_c}{b} + 1\right]$ | 17-18 7(a) |
| | **Lattice misfit** | $f = \dfrac{a_{layer} - a_{sub}}{a_{sub}}$ | 20-21 2(b) |
| | Critical thickness, estimate | $h_c \sim \dfrac{a_{sub}}{2\lvert f\rvert}$ | 20-21 2(b) |
| Epitaxy (8) | Si CVD | $\text{SiCl}_4 + 2\text{H}_2 \rightarrow \text{Si} + 4\text{HCl}$ | 18-19 1(d) |
| | GaAs chloride VPE | $6\text{GaCl} + \text{As}_4 + 3\text{H}_2 \rightarrow 6\text{GaAs} + 6\text{HCl}$ | 18-19 1(d) |

### Symbols:

$E_\Gamma$ = gap to the Γ valley (eV) ↩

$x$ = P fraction in $GaAs_{1-x}P_x$; mole fraction ↩

$E_X$ = gap to the X valleys (eV) ↩

$\lambda$ = wavelength of a photon of energy $E_g$ ↩

$a_L$ = layer ↩

$a_s$ = substrate ↩

$a$ = lattice constant (Å) ↩

$P$ = lattice constant or band gap of the alloy, from the four binaries ↩

$a_{layer}$ = layer lattice constant ↩

$a_{sub}$ = substrate lattice constant ↩



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## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **GaAsP band gaps** | | | | |
| 1 | 21-22 TT1 7(a) | 3 | theory repeats 17-18 1(c) | Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper *[sic: should be steeper]* and why? |
| 2 | 17-18 1(c) | 9 | theory | How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap. |
| **Quaternary alloys** | | | | |
| 3 | 20-21 2(d) | 7 | numerical | Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å. |
| 4 | 17-18 1(d) | 9 | numerical | For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system. |
| **Lattice matching** | | | | |
| 5 | 20-21 2(b) | 9 | theory | Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth. |
| 6 | 17-18 3(b) | 9 | theory | Discuss the significance of critical thickness in epitaxial growth. Draw enegy *[sic: should be energy]* band structure of Si and GaAs and show optical transitions in such semiconductors. |
| 7 | 17-18 TT1 7(a) | 8 | numerical | If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively. |
| **Epitaxy** | | | | |
| 8 | 18-19 1(d) | 8 | theory | Explain the process of vapor phase epitaxy. |
| **Isoelectronic traps** | | | | |
| 9 | 20-21 2(c) | 4 | theory | Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$. |

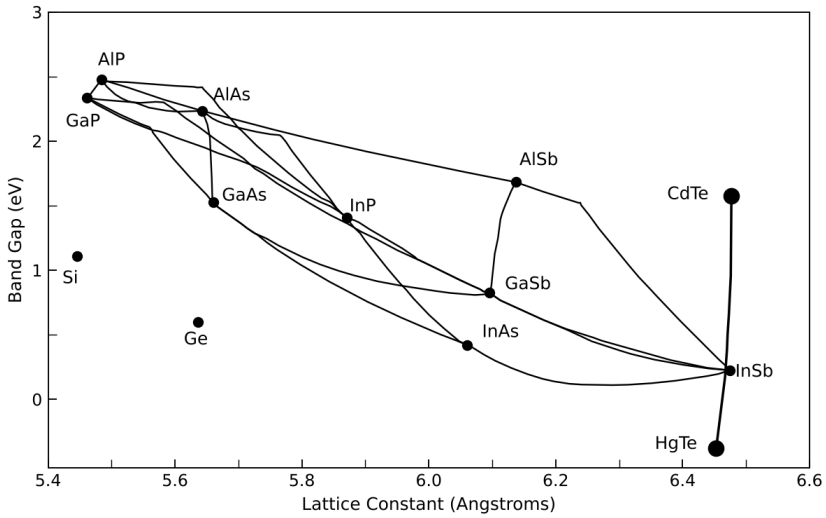



**Figure 1:** 20-21 2(d): Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)


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## 4.4. PN junction:

5 marks, 1 question, in 1 of 4 papers. Book: Neamen Ch. 7.

Most often theory (1 of 1). Subtopics: Graded junction.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Graded junction (5) | Net doping | $N_d - N_a = a\,x$ | 20-21 4(c) |
| | **Depletion width** | $W = \left[\frac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$ | 20-21 4(c) |
| | Junction capacitance | $C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi}+V_R)}\right]^{1/3}$ | 20-21 4(c) |

### Symbols:

$a$ = grading constant ($\text{cm}^{-4}$) ↩ $W$ = total depletion width (cm) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Graded junction** | | | | |
| 1 | 20-21 4(c) | 5 | theory | Describe how a linearly graded junction is formed. |

## 4.5. Metal-semiconductor contacts:

94 marks, 16 questions, in 4 of 4 papers. Book: Neamen Ch. 9; Sze Ch. 3.

Most often theory (7 of 16). Subtopics: Schottky junction; Ohmic contacts; Interface states; Barrier lowering.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Schottky junction (53) | **Schottky barrier** | $\phi_{B0} = \phi_m - \chi$ (n); $\phi_{Bp} = \chi + E_g/e - \phi_m$ (p) | 17-18 2(c), 17-18 3(a) |
| | Fermi level below $E_c$ | $\phi_n = V_t \ln(N_c/N_d)$ | 17-18 2(c) |
| | **Built-in potential** | $V_{bi} = \phi_{B0} - \phi_n$ | 18-19 1(c), 18-19 7(b), 17-18 2(c) |
| | **Depletion width** | $x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$ | 20-21 3(c), 17-18 2(c), 17-18 3(c) |
| | Peak field | $\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s}$ | 18-19 1(c), 17-18 2(c) |
| | **Barrier, n-type** | $\phi_{B0} = \phi_m - \chi$ | 21-22 7(c), 20-21 3(c), 18-19 1(c), 18-19 3(d), 18-19 7(b) |
| | **Barrier, p-type** | $\phi_{Bp} = \frac{E_g}{e} + \chi - \phi_m$ | 18-19 3(d) |
| | $E_c - E_F$ in n-type | $\phi_n = V_t \ln \frac{N_c}{N_d}$ | 18-19 1(c), 18-19 7(b) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **Depletion width** | $x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$ | 21-22 7(c), 18-19 1(c) |
| | **Built-in potential** | $V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$ | 20-21 3(c) |
| | **Maximum field** | $E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi}+V_R)}{x_n}$ | 20-21 3(c) |
| | Junction capacitance | $C' = \frac{\epsilon_s}{x_n} = \left[\frac{e\epsilon_s N_d}{2(V_{bi}+V_R)}\right]^{1/2}$ | 20-21 3(c) |
| | $E_c - E_F$ in n-type | $\phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$ | 21-22 7(c) |
| | **Built-in potential** | $V_{bi} = \phi_{B0} - \phi_n = \phi_m - \phi_s$ | 21-22 7(c) |
| | Poisson's equation | $\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}$ | 21-22 7(c) |
| | **Field, depletion region** | $E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x)$ | 21-22 7(c) |
| | Metal sheet charge | $Q'_m = -eN_d x_n$ | 21-22 7(c) |
| | **Maximum field** | $\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s} = \left[\frac{2eN_d V_{bi}}{\epsilon_s}\right]^{1/2}$ | 21-22 7(c) |
| Ohmic contacts (22) | Tunnelling contact | $R_c \propto \exp\left[\frac{2\sqrt{\epsilon_s m^*}}{\hbar}\frac{\phi_{Bn}}{\sqrt{N_d}}\right]$ | 17-18 3(c) |
| | **Specific contact resistance** | $R_c = \left(\frac{\partial J}{\partial V}\right)^{-1}_{V=0}$ | 18-19 3(a) |
| | Thermionic $R_c$ | $R_c = \frac{kT/e}{A^*T^2}\exp\left(\frac{e\phi_{Bn}}{kT}\right)$ | 18-19 3(a) |
| | Tunnelling $R_c$ | $R_c \propto \exp\left(\frac{2\sqrt{\epsilon_s m_n^*}}{\hbar}\cdot\frac{\phi_{Bn}}{\sqrt{N_d}}\right)$ | 18-19 3(a), 18-19 7(a) |
| | Ohmic contact (ideal) | $\phi_m < \phi_s$ (n-type), $\phi_m > \phi_s$ (p-type) | 20-21 5(d) |
| Interface states (5) | Interface states | $\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right),\ \gamma = \frac{\epsilon_i}{\epsilon_i + e^2\delta D_{it}}$ | 18-19 1(b) |
| Barrier lowering (14) | **Image-force lowering** | $\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$ | 20-21 3(d) |
| | Position of the peak | $x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$ | 20-21 3(d) |
| | Barrier with interface states | $\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right)$ | 20-21 4(b) |

## Symbols:

$V_R$ = reverse bias (V) ↩

$\phi_m$ = metal work function (V) ↩

$\chi$ = electron affinity (V) ↩

$N_c$ = effective density of states ($cm^{-3}$); conduction-band density of states ($cm^{-3}$) ↩

$\phi_n$ = depth of $E_F$ below $E_c$ (V) ↩

$C'$ = capacitance per unit area ($F/cm^2$) ↩

$N_d$ = donor concentration ($cm^{-3}$) ↩

$\phi_s$ = semiconductor work function (V) ↩

$A^*$ = Richardson constant ↩

$\phi_0$ = neutral level of the interface states above $E_v$ (V) ↩

$\delta$ = layer thickness ↩

$D_{it}$ = state density ↩

$\Delta\phi$ = barrier lowering (V) ↩

$E$ = field at the interface (V/cm) ↩

$x_m$ = distance of the barrier maximum from the metal (cm) ↩

$\gamma$ = 1 without interface states, 0 when the Fermi level is pinned ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Schottky junction** | | | | |
| 1 | 21-22 TT1 7(c) | 9 | derivation | A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions. |
| 2 | 20-21 3(a) | 5 | theory | Draw the ideal energy band diagram of a metal semiconductor junction under: (i) Reverse bias condition (ii) Forward bias condition |
| 3 | 20-21 3(c) | 8 | numerical | Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ $cm^{-3}$ at $T = 300$ K. Determine the (i) theoretical barrier height (ii) built-in potential barrier (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ $cm^{-3}$. |
| 4 | 18-19 1(c) | 6 | numerical | Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ $cm^{-3}$ at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$ |
| 5 | 18-19 3(d) | 0 | numerical | Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height. (i) Al on top of n-GaAs (ii) Cu on top of n-GaAs (iii) Ag on top of p-GaAs (iv) Au on top of p-GaAs |
| 6 | 18-19 TT1 7(b) | 6 | numerical repeats 18-19 1(c) | Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ $cm^{-3}$ at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$. |
| 7 | 17-18 2(c) | 10 | numerical | Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5 \times 10^{15}$ $cm^{-3}$. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. ($N_c$ = [unclear: $4.7 \times 10^{17}$] $cm^{-3}$, [unclear: $\epsilon_r$] 13.1, $\phi_m = 4.55$, $\chi = 4.07$) |
| 8 | 17-18 3(a) | 9 | diagram | Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors. |
| **Ohmic contacts** | | | | |
| 9 | 20-21 5(d) | 4 | theory | Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact. |
| 10 | 18-19 3(a) | 4 | theory | Define specific resistance. Explain which parameters affect specific resistance. |
| 11 | 18-19 3(c) | 3 | theory repeats 18-19 7(a) | Briefly explain the types of ohmic contacts? |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| 12 | 18-19 TT1 7(a) | 4 | theory | What is ohmic contact? Briefly explain the types of ohmic contacts. |
| 13 | 17-18 3(c) | 7 | theory | Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact? |
| **Interface states** | | | | |
| 14 | 18-19 1(b) | 5 | diagram | Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states. |
| **Barrier lowering** | | | | |
| 15 | 20-21 3(d) | 6 | numerical | Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm$^{-1}$. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height. |
| 16 | 20-21 4(b) | 8 | theory | Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction: (i) Image-force-induced lowering (ii) Interface states |

## 4.6. Heterojunctions:

81 marks, 13 questions, in 3 of 4 papers. Book: Neamen Ch. 9.

Most often theory (4 of 13). Subtopics: Band diagrams; HEMT; Fields and potentials; Resonant tunnelling.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Band diagrams (30) | **Anderson offsets** | $\Delta E_c = e(\chi_n - \chi_P),\ \Delta E_v = \Delta E_g - \Delta E_c$ | 18-19 2(a), 18-19 2(d), 17-18 2(a), 17-18 2(b) |
| | **Heterojunction $V_{bi}$** | $eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0} N_{vn}}{p_{n0} N_{vP}}\right)$ | 17-18 2(a) |
| | Depletion widths | $x_n = \left[\frac{2\epsilon_n \epsilon_P N_{aP} V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2},\ N_{dn} x_n = N_{aP} x_P$ | 17-18 2(a) |
| | AlGaAs gap | $E_g(x) \approx 1.424 + (E_{g,\text{AlAs}} - 1.424)x$ | 17-18 2(b) |
| | AlGaAs band gap | $E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x$ eV | 20-21 1(d), 20-21 7(d) |
| | **Band offsets (Anderson)** | $\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$ | 20-21 1(d), 20-21 7(d) |
| | Hole concentration | $p = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$ | 20-21 1(d), 20-21 7(d) |
| | **Built-in potential, pN** | $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{p_P\, N_{vN}}{p_N\, N_{vP}}\right)$ | 20-21 1(d), 20-21 7(d) |
| HEMT (21) | Quantum-well levels | $E_n = \frac{\hbar^2 \pi^2 n^2}{2m^* w^2}$ | 17-18 5(b) |
| | **2DEG density** | $n_s = \frac{\epsilon_N}{e(d + \Delta d)}(V_g - V_{off})$ | 18-19 6(b), 17-18 5(b), 17-18 5(d) |
| | **HEMT off voltage** | $V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2},\ V_{p2} = \frac{eN_d {d_d}^2}{2\epsilon_N}$ | 17-18 5(d) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | HEMT current | $I_D(\text{sat}) = \dfrac{\epsilon_N W}{d+\Delta d}(V_g - V_{off} - V_0)v_{sat}$ | 17-18 5(b) |
| | HEMT off voltage | $V_{off} = \phi_B - \dfrac{\Delta E_c}{e} - \dfrac{eN_d d_d^2}{2\epsilon_N}$ | 18-19 6(b) |
| Fields and potentials (24) | **Built-in potential (nP)** | $eV_{bi} = \Delta E_v + kT\ln\left(\dfrac{p_{P0}}{p_{n0}}\cdot\dfrac{N_{vn}}{N_{vP}}\right)$ | 18-19 2(d) |
| | Charge balance | $N_{dn}x_n = N_{aP}x_P$ | 18-19 2(b) |
| | Potential drops | $V_{bin} = \dfrac{eN_{dn}x_n^2}{2\epsilon_n},\ V_{biP} = \dfrac{eN_{aP}x_P^2}{2\epsilon_P}$ | 18-19 2(b) |
| | **Depletion widths** | $x_n = \left[\dfrac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn}+\epsilon_P N_{aP})}\right]^{1/2},\ x_P = \left[\dfrac{2\epsilon_n\epsilon_P N_{dn}V_{bi}}{eN_{aP}(\epsilon_n N_{dn}+\epsilon_P N_{aP})}\right]^{1/2}$ | 18-19 2(b), 18-19 2(d) |
| | Total width | $W = \left[\dfrac{2\epsilon_n\epsilon_P(N_{dn}+N_{aP})^2 V_{bi}}{eN_{dn}N_{aP}(\epsilon_n N_{dn}+\epsilon_P N_{aP})}\right]^{1/2}$ | 18-19 2(b) |
| | Field, N side | $E_1(x) = \dfrac{eN_{d1}}{\epsilon_1}(x+x_n)$ | 20-21 4(a) |
| | Field, P side | $E_2(x) = \dfrac{eN_{a2}}{\epsilon_2}(x_p - x)$ | 20-21 4(a) |
| | **Interface condition** | $\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1}x_n = N_{a2}x_p$ | 20-21 4(a) |
| | Potential split | $V_{bi1} = \dfrac{eN_{d1}x_n^2}{2\epsilon_1},\ V_{bi2} = \dfrac{eN_{a2}x_p^2}{2\epsilon_2},\ V_{bi} = V_{bi1} + V_{bi2}$ | 20-21 4(a) |
| | Depletion width, N side | $x_n = \left[\dfrac{2\epsilon_1\epsilon_2 N_{a2}V_{bi}}{eN_{d1}(\epsilon_1 N_{d1}+\epsilon_2 N_{a2})}\right]^{1/2}$ | 20-21 4(a) |
| Resonant tunnelling (6) | RTD levels | $E_n \approx \dfrac{\hbar^2}{2m^*}\left(\dfrac{n\pi}{L_w}\right)^2$ | 18-19 3(b) |

## Symbols:

$\chi_n$, $\chi_w$ = electron affinities of the narrow- and wide-gap materials (V) ↩

$\Delta E_c$, $\Delta E_v$ = conduction- and valence-band offsets (eV) ↩

$N_v$ = valence-band effective density of states ($\text{cm}^{-3}$) ↩

$p_P$, $p_N$ = hole concentrations in the P and N regions ↩

$N_{vP}$, $N_{vN}$ = their valence-band densities of states ↩

$d$ = AlGaAs thickness, $\Delta d \approx 80$ Å ↩

$d_d$ = doped AlGaAs thickness ↩

$N_{d1}$ = donors ↩

$\epsilon_1$ = permittivity of side 1 ↩

$N_{a2}$ = acceptors ↩

$\epsilon_2$ = permittivity of side 2 ↩

$V_{bi1}$, $V_{bi2}$ = parts of $V_{bi}$ dropped on each side ↩

$L_w$ = well width ↩

## Questions:


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Band diagrams** | | | | |
| 1 | 20-21 1(d) | 8 | derivation | For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following: (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P\, N_{VN}}{P_N\, N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped $N^+$ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ $P_p$ = hole concentration in P type region, $P_N$ = hole concentration in N type region, $N_{VN}$ and $N_{VP}$ are density of states of electron and hole respectively. |
| 2 | 20-21 TT1 7(d) | 10 | derivation repeats 20-21 1(d) | For a GaAs-$Al_xGa_{(1-x)}As$ heterojunction, determine the following: (a) Energy-Band Diagram: (i) before contact (ii) After contact (b) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_p\, N_{VN}}{P_N\, N_{VP}}\right)$. Given, GaAs is doped p type, $Al_xGa_{(1-x)}As$ is doped N+ type. Assume: x=0.25 & $\Delta E_c = \frac{1}{2}\Delta E_g$. $P_p$ = Hole concentration in P type region, $P_N$ = hole concentration in N type region, $N_{VN}$ and $N_{VP}$ are density of states of electron and hole, respectively. $Hole\ concentration, P = N_V \exp\left(\frac{-(E_F - E_V)}{kT}\right)$ |
| 3 | 18-19 2(a) | 5 | diagram | Draw the band diagrams of a pN and a nN hetero-junction. |
| 4 | 17-18 2(a) | 7 | numerical | What are the assumptions made in Anderson's model? Determine $\Delta E_C$, $\Delta E_V$ and $V_{bi}$ for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, $n_i$ = [unclear: $2.5 \times 10^{13}$] $cm^{-3}$ (for Ge), $N_d = 1.2 \times 10^{16}$ $cm^{-3}$, and $N_a = 1 \times 10^{16}$ $cm^{-3}$. |
| 5 | 17-18 2(b) | 0 | diagram | Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for: (i) [unclear: $N^+$]-AlGaAs, intrinsic GaAs. (ii) $P^+$-AlGaAs, n-GaAs. |
| **HEMT** | | | | |
| 6 | 18-19 6(a) | 4 | theory | What are the advantages of High Electron Mobility Transistor (HEMT)? |
| 7 | 18-19 6(b) | 7 | theory repeats 17-18 5(b) | With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates. |
| 8 | 17-18 5(b) | 5 | theory | What is 2DEG? With quantum well structure describe the operation of HEMT. |
| 9 | 17-18 5(d) | 5 | numerical | Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N - Al_{0.3}Ga_{0.7}As$ layer doped to $10^{18}$ $cm^{-3}$ and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2. |
| **Fields and potentials** | | | | |
| 10 | 20-21 4(a) | 12 | derivation | A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions. |
| 11 | 18-19 2(b) | 7 | derivation | Derive the equation for the depletion region width of a hetero-junction. |
| 12 | 18-19 2(d) | 5 | numerical | Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, $N_{vn}$ = [unclear: 6 ×] $10^{18}$ cm$^{-3}$, $N_d = 10^{16}$ cm$^{-3}$, $n_i = 2.4 \times 10^{13}$ cm$^{-3}$ and $\varepsilon_n = 16$. For P-type GaAs electron affinity 4.07, $N_{vP} = 7 \times 10^{18}$ cm$^{-3}$, $N_a = 10^{16}$ cm$^{-3}$ and $\varepsilon_P = 13.1$ |
| **Resonant tunnelling** | | | | |
| 13 | 18-19 3(b) | 6 | theory | With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode. |


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## 4.7. MOS capacitor:

22 marks, 4 questions, in 1 of 4 papers. Book: Neamen Ch. 10.

Most often diagram (2 of 4). Subtopics: MOS capacitor; Threshold voltage.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| MOS capacitor (11) | Fermi potential | $\phi_{fp} = V_t \ln \frac{N_a}{n_i}$ | 18-19 4(c), 18-19 8(b) |
| | **Maximum depletion width** | $x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$ | 18-19 4(c), 18-19 8(b) |
| | Oxide capacitance | $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$ | 18-19 4(c) |
| | Depletion C | $C' = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d}$ | 18-19 4(c) |
| | Flat-band C | $C'_{FB} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)L_D}$ | 18-19 4(c) |
| Threshold voltage (11) | **Flat-band voltage** | $V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$ | 18-19 4(d) |
| | **Threshold voltage** | $V_{TN} = V_{FB} + \frac{\lvert Q'_{SD}(\max)\rvert}{C_{ox}} + 2\phi_{fp}$ | 18-19 4(d), 18-19 8(b) |

### Symbols:

$Q'_{ss}$ = oxide charge (C/cm$^2$) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **MOS capacitor** | | | | |
| 1 | 18-19 4(c) | 6 | diagram | For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve. |
| 2 | 18-19 TT2 8(a) | 5 | diagram | Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type *[sic: should be n-type]* substrate. |
| **Threshold voltage** | | | | |
| 3 | 18-19 4(d) | 6 | derivation | Show the relation between Flat band voltage and threshold voltage of a MOS device. |
| 4 | 18-19 TT2 8(b) | 5 | numerical | A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}\ cm^{-3}$ It has $Q'_{ss} = 10^{10}\ cm^{-2}$ and $t_{ox} = 30\ nm$ Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}$ cm$^{-3}$, $N_c = 2.8 \times 10^{19}$ cm$^{-3}$, $n_i = 1.5 \times 10^{10}$ cm$^{-3}$ and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V. |

## 4.8. Bipolar transistors:

108 marks, 16 questions, in 3 of 4 papers. Book: Neamen Ch. 12; Sze Ch. 5.

Most often theory (7 of 16). Subtopics: Ebers-Moll; Bandgap narrowing; Breakdown; TEBT; HBT; Base minority carriers; Injection efficiency; Non-ideal effects; Gummel-Poon; High-level injection.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Ebers-Moll (22) | **Ebers–Moll** | $I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R$ | 17-18 4(a) |
| | Reciprocity | $\alpha_F I_{ES} = \alpha_R I_{CS}$ | 20-21 6(a), 17-18 4(a) |
| | **Ebers–Moll** | $I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ \alpha_F I_{ES} = \alpha_R I_{CS}$ | 18-19 6(c) |
| | **Ebers–Moll currents** | $I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ I_B = I_E - I_C$ | 20-21 6(a) |
| | Diode currents | $I_F = I_{ES}\left[e^{V_{BE}/V_t} - 1\right],\ I_R = I_{CS}\left[e^{V_{BC}/V_t} - 1\right]$ | 20-21 6(a) |
| | **Saturation voltage** | $V_{CE}(sat) = V_t \ln\left[\frac{I_C(1-\alpha_R) + I_B}{\alpha_F I_B - (1-\alpha_F) I_C} \cdot \frac{\alpha_F}{\alpha_R}\right]$ | 20-21 6(a) |
| Bandgap narrowing (8) | Emitter minority holes | $p_{E0} = \frac{n_i^2}{N_E}$ | 17-18 4(d) |
| | **Bandgap narrowing** | $p'_{E0} = \frac{n_i^2}{N_E} \exp\left(\frac{\Delta E_g}{kT}\right)$ | 17-18 4(b), 17-18 4(d) |
| Breakdown (9) | Avalanche multiplication | $M = \frac{1}{1 - (V/BV_{CBO})^n}$ | 17-18 4(c) |
| | **Open-base breakdown** | $BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$ | 17-18 4(c) |
| TEBT (4) | Injection efficiency | $\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}}$ | 17-18 5(a) |
| | Tunnelling probability | $T \approx \exp\left(-2d\sqrt{2m^* V_0}/\hbar\right)$ | 17-18 5(a) |
| HBT (24) | **Kirk effect** | $n_c = \frac{J_C}{ev_{sat}};\ J_K = ev_{sat}\left[N_C + \frac{2\epsilon_s(V_{CB} + V_{bi})}{eW_C^2}\right]$ | 17-18 5(c) |
| | HBT injection ratio | $\frac{J_{pE}}{J_{nE}} \propto \exp\left(-\frac{\Delta E_g}{kT}\right)$ (graded); $\exp(-\Delta E_v/kT)$ (abrupt) | 17-18 5(c) |
| | **HBT injection ratio** | $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nB}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ | 18-19 4(b) |
| Base minority carriers (8) | **Base minority electrons** | $\delta n_B = \frac{n_{B0}(e^{eV_{BE}/kT} - 1)\sinh\frac{x_B - x}{L_B} - n_{B0}\sinh\frac{x}{L_B}}{\sinh(x_B/L_B)}$ | 18-19 2(c) |
| Injection efficiency (12) | **Injection efficiency** | $\gamma = \left[1 + \frac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \frac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}\right]^{-1}$ | 18-19 5(a), 18-19 5(d) |
| | Base transport factor | $\alpha_T = \frac{1}{\cosh(x_B/L_B)}$ | 18-19 5(d) |
| | **Gain** | $\alpha = \gamma\alpha_T\delta,\ \beta = \frac{\alpha}{1-\alpha}$ | 18-19 5(d) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Non-ideal effects (6) | Early effect | $\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE}+V_A}$, $r_o = \frac{V_A}{I_C}$ | 18-19 5(b) |
| Gummel-Poon (7) | Gummel–Poon current | $J_n = \frac{-eD_n n_i^2 e^{eV_{BE}/kT}}{\int_0^{x_B} p\,dx}$ | 18-19 5(c) |
| | Normalised base charge | $q_b = \frac{q_1}{2}(1+\sqrt{1+4q_2})$ | 18-19 5(c) |

## Symbols:

$\alpha_F$, $\alpha_R$ = forward and reverse common-base gains ↩

$I_{ES}$, $I_{CS}$ = reverse saturation currents of the B–E and B–C junctions (A) ↩

$n$ = empirical, 3 to 6 ↩

$d$ = barrier thickness ↩

$\delta$ = recombination factor ↩

$V_A$ = Early voltage ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Ebers-Moll** | | | | |
| 1 | 20-21 6(a) | 9 | derivation | Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model. |
| 2 | 18-19 6(c) | 5 | diagram repeats 17-18 4(a) | Sketch the equivalent circuit for Ebers-Moll model as used in BJT. |
| 3 | 17-18 4(a) | 8 | theory | From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates. |
| **Bandgap narrowing** | | | | |
| 4 | 17-18 4(b) | 3 | theory | What is bandgap narrowing? |
| 5 | 17-18 4(d) | 5 | numerical | Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from $10^{18}$ to $10^{19}$ $cm^{-3}$. Determine the new value of $p'_{E0}$ and determine the ratio $p'_{E0}/p_{E0}$. |
| **Breakdown** | | | | |
| 6 | 17-18 4(c) | 9 | derivation | Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage." |
| **TEBT** | | | | |
| 7 | 17-18 5(a) | 4 | theory | With neat sketch, describe basic operation mechanism of TEBT. |
| **HBT** | | | | |
| 8 | 18-19 4(a) | 6 | diagram | Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile |
| 9 | 18-19 4(b) | 7 | derivation | Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of $J_{nE}$ and $J_{pE}$ can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ *[sic: should be $D_{nB}$].* |
| 10 | 17-18 5(c) | 11 | theory | With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters. |
| **Base minority carriers** | | | | |
| 11 | 18-19 2(c) | 8 | derivation | For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor |
| **Injection efficiency** | | | | |
| 12 | 18-19 5(a) | 7 | derivation | Derive the equation for emitter injection factor of a bipolar junction transistor. |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| 13 | 18-19 5(d) | 5 | numerical | Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23$ cm$^2$/s, $D_E = 8$ cm$^2$/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm$^{-3}$, and $x_E = 0.35$ $\mu$m. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ $\mu$m can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, $N_E$, to meet this specification. (b) Using the results of part (a), what are the values of $\alpha_T$ and $\gamma$? |
| **Non-ideal effects** | | | | |
| 14 | 18-19 5(b) | 6 | theory | Briefly explain the following non-ideal effects of a bipolar junction transistor. (i) Early effect (ii) Current Crowding |
| **Gummel-Poon** | | | | |
| 15 | 18-19 5(c) | 7 | theory | Describe the Gummel-Poon model of a bipolar junction transistor. |
| **High-level injection** | | | | |
| 16 | 20-21 6(b) | 8 | theory | Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved? |

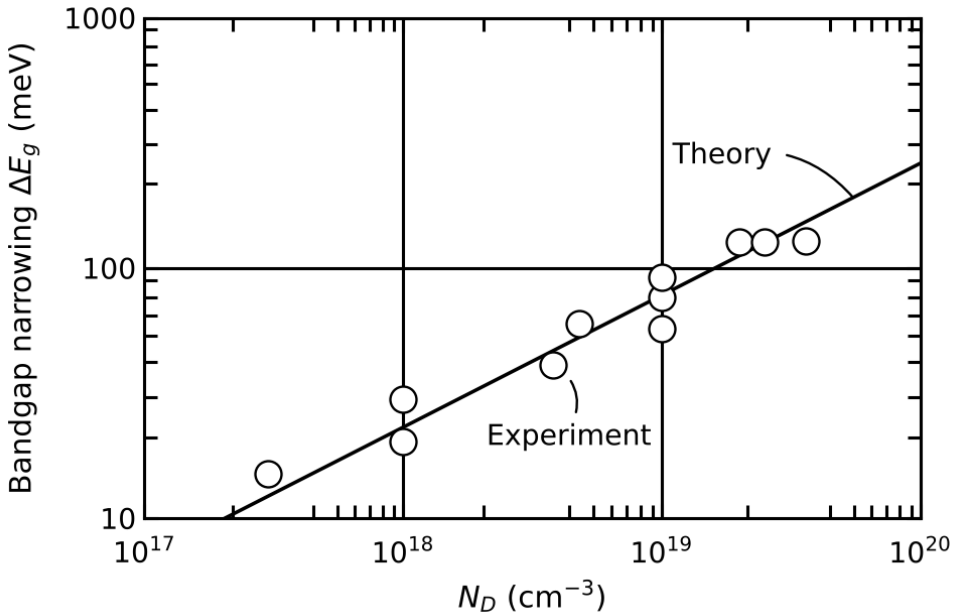



**Figure 2:** 17-18 4(d): Figure 1 of the paper (redrawn)


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## 4.9. JFETs:

63 marks, 10 questions, in 3 of 4 papers. Book: Neamen Ch. 13; Sze Ch. 7.

Most often numerical (4 of 10). Subtopics: Channel profiles; Pinch-off; Transconductance; Cut-off frequency.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Channel profiles (14) | Depletion thickness | $h = \left[\dfrac{2\epsilon_s(V_{bi} - V_{gs} + V(x))}{eN_d}\right]^{1/2}$ | 17-18 6(a), 17-18 6(b) |
| | Saturation drain voltage | $V_{d,sat} = V_g - V_p$ | 17-18 6(a) |
| | Saturation drain voltage | $V_{DS}(sat) = V_{p0} - (V_{bi} - V_{GS})$ | 20-21 5(a) |
| Pinch-off (31) | **Internal pinch-off** | $V_{p0} = \dfrac{ea^2N_d}{2\epsilon_s}$ ($N_a$ for p channel) | 17-18 6(b), 17-18 6(c) |
| | **Pinch-off voltage** | $V_p = V_{bi} - V_{p0}$ (n); $V_p = V_{p0} - V_{bi}$ (p) | 17-18 6(c) |
| | pn built-in potential | $V_{bi} = V_t \ln\left(\dfrac{N_aN_d}{n_i^2}\right)$ | 17-18 6(c) |
| | **MESFET threshold** | $V_T = V_{bi} - V_{p0}$, $V_{bi} = \phi_{Bn} - \phi_n$ | 18-19 6(e) |
| | pn built-in potential | $V_{bi} = V_t \ln \dfrac{N_aN_d}{n_i^2}$ | 20-21 5(c) |
| | **Space-charge width** | $h = \left[\dfrac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$ | 20-21 5(b) |
| | Pinch-off voltage | $V_p = V_{bi} - V_{p0}$ | 20-21 5(b) |
| Transconductance (14) | Square law | $I_D = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_p}\right)^2$ | 17-18 6(d) |
| | **Transconductance** | $g_m = \left.\dfrac{\partial I_D}{\partial V_{GS}}\right\|_{V_{DS}}$; $g_{ms} = -\dfrac{2I_{DSS}}{V_p}\left(1 - \dfrac{V_{GS}}{V_p}\right)$ | 17-18 6(d) |
| | Source resistance | $g_m' = \dfrac{g_m}{1 + g_m r_s}$ | 17-18 6(d) |
| | **Pinch-off current** | $I_{P1} = \dfrac{\mu_n(eN_d)^2Wa^3}{6\epsilon_s L}$ | 20-21 6(c) |
| | Saturation current | $I_{D1}(sat) = I_{P1}\left[1 - 3\left(\dfrac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \dfrac{2}{3}\sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$ | 20-21 6(c) |
| | **Transconductance** | $g_{ms} = \dfrac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)$ | 20-21 6(c) |
| Cut-off frequency (4) | **Cut-off frequency** | $f_T = \dfrac{g_m}{2\pi C_G}$; $f_T(\max) = \dfrac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$ | 18-19 6(d) |
| | Internal pinch-off | $V_{p0} = \dfrac{ea^2N_d}{2\epsilon_s}$ | 20-21 5(b), 18-19 6(d), 18-19 6(e) |


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## Symbols:

$a$ = metallurgical channel thickness (cm)

$h$ = depletion width into the channel (cm) ↩

$L$ = channel length (cm); gate length

$N_a$ = gate acceptors ↩

$V_{GS}$ = gate–source voltage (V) ↩

$\mu_n$ = electron mobility ($cm^2/V \cdot s$) ↩

$N_d$ = channel donors ($cm^{-3}$) ↩

$W$ = channel width

$g_{ms}$ = saturation transconductance (S) ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Channel profiles** | | | | |
| 1 | 20-21 5(a) | 6 | theory repeats 17-18 6(a) | Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where $V_p$, $V_g$, and $V_d$ are pinch-off, gate, and drain voltage respectively) (i) $V_g > V_p$ and $V_d < V_{d,sat}$ (ii) $V_g > V_p$ and $V_d = V_{d,sat}$ (iii) $V_g > V_p$ and $V_d > V_{d,sat}$ |
| 2 | 17-18 6(a) | 8 | diagram | Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where $V_p$, $V_g$, and $V_d$ are pinch-off, gate, and drain voltages respectively) (i) $V_g > V_p$ and $V_d = 0V$ (ii) $V_g > V_p$ and $V_d < V_{d,sat}$ (iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$ (iv) $V_g > V_p$ and $V_d > V_{d,sat}$ |
| **Pinch-off** | | | | |
| 3 | 20-21 5(b) | 8 | derivation repeats 17-18 6(b) | Derive the formula of internal pinch-off voltage in an n-channel JFET. |
| 4 | 20-21 5(c) | 7 | numerical | Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ $cm^{-3}$, $N_d = 3 \times 10^{15}$ $cm^{-3}$, $\epsilon_s = 13.1$, and $a = 0.70$ $\mu$m. Determine the forward-bias gate voltage required to open a channel region that is 0.10 $\mu$m thick with zero drain voltage. |
| 5 | 18-19 6(e) | 5 | numerical | Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ $cm^{-3}$. |
| 6 | 17-18 6(b) | 6 | derivation | In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \frac{ea^2 N_d}{2\epsilon_s}$ |
| 7 | 17-18 6(c) | 5 | numerical | The $n^+p$ junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}$ $cm^{-3}$ and $N_a = 10^{16}$ $cm^{-3}$. The metallurgical channel thickness is $a = 0.40$ $\mu m$. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET. |
| **Transconductance** | | | | |
| 8 | 20-21 6(c) | 8 | numerical | Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ $cm^{-3}$, $N_d = 1 \times 10^{16}$ $cm^{-3}$, $a = 0.75$ $\mu$m, $L = 10$ $\mu$m, $W = 30$ $\mu$m, and $\mu_n = 1000$ $cm^2/(V\ s)$ (i) Calculate the maximum current in an n-channel JFET. (ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V. |
| 9 | 17-18 6(d) | 6 | theory | Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET. |
| **Cut-off frequency** | | | | |
| 10 | 18-19 6(d) | 4 | derivation | Show that maximum cut off frequency of a MESFET is related to doping concentration. |


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# 5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| **Band structure and materials** | | | | |
| 1 | 21-22 TT1 7(b) | 8 | Density of states | $D(E)$ counts states per unit energy, from the allowed $k$ points ($2\pi/L$ grid, 2 per point) and $E = \hbar^2k^2/2m^*$. 3D: shell $4\pi k^2 dk$, $D \propto \sqrt{E - E_c}$. 2D well: ring $2\pi k\,dk$, constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: two segments, $D \propto 1/\sqrt{E - E_n}$ spikes. 0D dot: delta functions. In general $D \propto (E - E_n)^{d/2-1}$. |
| 2 | 20-21 2(a) | 5 | Materials | Near-IR (0.75 to 2 $\mu$m) uses III-V compounds: GaAs/AlGaAs (0.8 to 0.9 $\mu$m), InGaAsP and InGaAs on InP (1.3 and 1.55 $\mu$m, the fibre windows), Ge or InGaAs detectors. Reasons: band gaps 0.6 to 1.65 eV give $\lambda = 1.24/E_g$ in the near-IR; direct gaps give efficient emission; alloys tune the gap while lattice-matched to InP or GaAs. |
| 3 | 20-21 3(b) | 6 | Density of states | $D(E)$ counts states per unit energy. 3D: $D \propto \sqrt{E - E_c}$. 2D quantum well: constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: $D \propto 1/\sqrt{E - E_n}$ spikes at each subband edge. 0D dot: delta functions. Confinement squeezes states into sharper features. |
| 4 | 17-18 1(a) | 3 | Crystal structure | Single crystal: long-range order throughout (Si wafer). Polycrystalline: ordered grains of different orientation separated by grain boundaries (polysilicon). Amorphous: short-range order only (a-Si:H, $SiO_2$). |
| **High-field transport** | | | | |
| 5 | 20-21 1(a) | 6 | Mobility | GaAs: $\mu_n = 8500$ against 1350 cm$^2$/V s for Si. $\mu = e\tau/m^*$: GaAs has one sharply curved direct $\Gamma$ valley with $m^* = 0.067\,m_0$; Si has six flat X valleys (conductivity mass $0.26\,m_0$) and intervalley scattering among them; GaAs also has weaker phonon scattering in the $\Gamma$ valley. |
| 6 | 20-21 1(b) | 5 | Gunn effect | Velocity saturation: hot carriers lose their extra energy to optical phonons, so $v_d$ levels off near $10^7$ cm/s (Si above about $3 \times 10^4$ V/cm). Gunn effect: in n-GaAs above about 3.2 kV/cm electrons transfer from the light $\Gamma$ valley to the heavy L valley, $v_d$ falls (negative differential mobility), and high-field domains form and travel, giving microwave current pulses. |
| 7 | 20-21 TT1 7(a) | 3 | Mobility | Si 1350 against GaAs 8500 cm$^2$/V s. $\mu = e\tau/m^*$: Si electrons sit in six flat X valleys (conductivity mass $0.26\,m_0$), GaAs electrons in one sharp $\Gamma$ valley ($0.067\,m_0$), 3.9 times lighter; Si also has intervalley scattering among its X valleys, so $\tau$ is shorter (0.20 against 0.32 ps). |
| 8 | 20-21 TT1 7(b) | 3 | Gunn effect | In n-GaAs above about 3.2 kV/cm hot electrons transfer from the light $\Gamma$ valley to the heavy L valleys 0.31 eV higher; $v_d$ falls as $E$ rises (negative differential mobility); high-field domains form at the cathode and drift to the anode, giving current pulses at $f = v_d/L$ (10 GHz for 10 $\mu$m). |
| 9 | 17-18 TT1 7(b) | 12 | Mobility | $m^* = \hbar^2/(d^2E/dk^2)$: GaAs single sharp $\Gamma$ valley, $0.067\,m_0$; Si six flat X ellipsoids, $m^*_{cn} = 0.26\,m_0$. GaAs mobility higher (8500 vs 1350 cm$^2$/V $\cdot$ s, $\mu = e\tau/m^*$). Field: constant at low field; Si saturates at $10^7$ cm/s; GaAs peaks at $2 \times 10^7$ cm/s at 3.2 kV/cm, negative differential mobility (L valley), saturates. |
| **Alloys and strain** | | | | |
| 10 | 21-22 TT1 7(a) | 3 | GaAsP band gaps | Both gaps of $GaAs_{1-x}P_x$ rise with $x$: $E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ eV; they cross at $x_c \approx 0.45$–$0.48$ (direct below, indirect above). The $\Gamma$ gap is steeper (1.33 against 0.36 eV from GaAs to GaP): the s-like antibonding $\Gamma$ state rises sharply as the smaller P shortens the bonds (large deformation potential); the p-like X state hardly moves. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 11 | 20-21 2(b) | 9 | Lattice matching | Matching ($f = (a_L - a_S)/a_S \approx 0$) avoids misfit and threading dislocations, which are non-radiative recombination and scattering centres. A thin mismatched layer grows strained (pseudomorphic, tetragonal distortion) up to a critical thickness $h_c$ (Matthews–Blakeslee), beyond which dislocations relax it; strain also shifts band edges and can be used on purpose. |
| 12 | 20-21 2(c) | 4 | Isoelectronic traps | $GaAs_{1-x}P_x$ is direct for $x < 0.45$ and indirect above. Nitrogen on a P site is isoelectronic and forms a trap just below $E_c$; the electron is localised in space, so spread in $k$, and can recombine with a hole without a phonon; this makes indirect GaAsP:N (red-orange, yellow, green) LEDs efficient. |
| 13 | 18-19 1(d) | 8 | Epitaxy | Single-crystal layer grown on a heated substrate from reacting gases (CVD). Si: $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$. GaAs chloride VPE: $AsCl_3 + H_2$ over a Ga boat (850 °C) gives GaCl and $As_4$, which deposit GaAs at 750 °C; MOVPE uses $Ga(CH_3)_3 + AsH_3$. Thickness by time and flow, doping by dopant gases, composition by mixing sources. |
| 14 | 17-18 1(c) | 9 | GaAsP band gaps | $E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ eV; direct below $x = 0.45$, indirect above. Without N: efficient emission only below 0.45. With the N isoelectronic trap: electron localised in space, spread in $k$, bound exciton, phonon-free recombination: efficient orange, yellow, green LEDs above 0.45. |
| 15 | 17-18 3(b) | 9 | Lattice matching | Critical thickness: largest thickness a mismatched layer grows strained without misfit dislocations, $h_c \approx a_s/2\lvert f\rvert$; above it dislocations lower lifetime, efficiency and mobility. GaAs direct gap: vertical transitions, strong absorption and emission. Si indirect: phonon needed, weak absorption near the edge, very weak emission. |
| **PN junction** | | | | |
| 16 | 20-21 4(c) | 5 | Graded junction | Diffuse acceptors into a uniformly doped n-type wafer (or grade the doping during epitaxy); near the depth where $N_a = N_d$ the net doping is $a(x - x_j)$: linear. Then $\rho = eax$, the field is parabolic, $W = [12\epsilon_s(V_{bi} + V_R)/(ea)]^{1/3}$ and $C' \propto (V_{bi} + V_R)^{-1/3}$. |
| **Metal-semiconductor contacts** | | | | |
| 17 | 20-21 3(a) | 5 | Schottky junction | Ideal metal/n-type contact: the barrier from the metal stays $e\phi_{B0} = e(\phi_m - \chi)$. Reverse bias raises the barrier seen from the semiconductor to $e(V_{bi} + V_R)$, widens the depletion layer, tiny current. Forward bias lowers it to $e(V_{bi} - V_a)$, narrows the layer, electrons flow into the metal. |
| 18 | 20-21 4(b) | 8 | Barrier lowering | (i) The image charge of an electron lowers the barrier by $\Delta\phi = \sqrt{eE/4\pi\epsilon_s}$ at $x_m = \sqrt{e/16\pi\epsilon_s E}$; $E$ grows with reverse bias, so the reverse current rises with $V_R$. (ii) Interface states with a neutral level $e\phi_0$ pin the surface Fermi level, so the barrier depends only weakly on the metal work function (Bardeen limit). |
| 19 | 20-21 5(d) | 4 | Ohmic contacts | With $\phi_m < \phi_s$ on n-type, electrons flow from the metal into the semiconductor and form an accumulation layer; the bands bend down by $e(\phi_s - \phi_m)$; electrons meet no barrier either way, so the I-V is linear with low resistance: an ohmic contact. |
| 20 | 18-19 3(a) | 4 | Ohmic contacts | Specific contact resistance $R_c = (\partial J/\partial V)^{-1}_{V=0}$ ($\Omega$ cm$^2$). Thermionic: $R_c = (kT/e)e^{e\phi_{Bn}/kT}/(A^*T^2)$, exponential in the barrier. Tunnelling ($N_d \gtrsim 10^{19}$): $R_c \propto \exp\left[2\sqrt{\epsilon_s m^*}\phi_{Bn}/(\hbar\sqrt{N_d})\right]$. Also temperature, $m^*$, $\epsilon_s$, interface quality. |
| 21 | 18-19 3(c) | 3 | Ohmic contacts | Barrier-free contact: $\phi_m < \phi_s$ on n-type or $\phi_m > \phi_s$ on p-type, majority carriers accumulate. Tunnelling contact: heavy surface doping makes the barrier a few nm thick, carriers tunnel through. |
| 22 | 18-19 TT1 7(a) | 4 | Ohmic contacts | Ohmic: linear symmetric I-V, small $R_c$. Types: barrier-free ($\phi_m < \phi_s$ on n-type, $\phi_m > \phi_s$ on p-type, accumulation) and tunnelling (heavy surface doping, thin barrier, $R_c$ falls with $\sqrt{N_d}$). |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 23 | 17-18 3(c) | 7 | Ohmic contacts | Metal on $n^+$ ($\gtrsim 10^{19}$ cm$^{-3}$): depletion width $\propto N_d^{-1/2}$ falls to a few nm; electrons tunnel (field emission) through the barrier in both directions; linear I–V, small $R_c$: ohmic. |
| **Heterojunctions** | | | | |
| 24 | 18-19 3(b) | 6 | Resonant tunnelling | Double barrier with a quantum well: levels $E_1$, $E_2$. Bias lowers the levels; current peaks when $E_1$ lies in the band of emitter electrons (resonant tunnelling) and falls when it drops below the emitter $E_c$ (NDR); $E_2$ gives a second peak and a second NDR region. |
| 25 | 18-19 6(a) | 4 | HEMT | 2DEG in undoped GaAs separated from its donors (modulation doping): high mobility (8500 cm$^2$/V s at 300 K, far more at 77 K), high speed ($f_T$ above 100 GHz), high transconductance (about 250 mS/mm), lowest noise, high sheet density, low-voltage operation. |
| 26 | 18-19 6(b) | 7 | HEMT | N-AlGaAs / spacer / undoped GaAs: electrons drop by $\Delta E_c$ into a triangular well, a 2DEG; the Schottky gate sets $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$ with $V_{off} = \phi_B - \Delta E_c/e - eN_d d_d^2/2\epsilon_N$; negative $V_g$ empties the well, positive fills it; $I_D$ saturates at $v_{sat}$. |
| 27 | 17-18 5(b) | 5 | HEMT | 2DEG: electrons quantised across a narrow well, free in the plane. N-AlGaAs donors give electrons to the triangular well in undoped GaAs (modulation doping, high mobility); gate sets $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$; negative $V_g$ empties the well, positive fills it until the AlGaAs conducts. |
| **Bipolar transistors** | | | | |
| 28 | 20-21 6(b) | 8 | High-level injection | High-level injection: injected electrons in the base approach $N_B$, extra holes come in for neutrality, effective base doping rises, injection efficiency falls, $I_C \propto e^{V_{BE}/2V_t}$, $\beta$ falls at high current. Current crowding: the lateral base current's $IR$ drop makes $V_{BE}$ largest at the emitter edges, so current crowds there (hot spots). Cure: narrow interdigitated emitter fingers, more base contacts, ballast resistors. |
| 29 | 18-19 5(b) | 6 | Non-ideal effects | Early effect: B-C depletion widens with $V_{CB}$, neutral base narrows, $I_C$ rises, curves extrapolate to $-V_A$, $r_o = V_A/I_C$; punch-through. Current crowding: lateral base $IR$ drop makes $V_{BE}$ largest at the emitter edges; current crowds there; cure: interdigitated narrow emitters, ballast. |
| 30 | 18-19 5(c) | 7 | Gummel-Poon | Charge-control model: $J_n = -eD_n n_i^2 e^{eV_{BE}/kT} / \int_0^{x_B} p\,dx$ (base Gummel number); $I_C = (I_S/q_b)(e^{V_{BE}/V_t} - e^{V_{BC}/V_t})$, $q_b = (q_1/2)(1 + \sqrt{1 + 4q_2})$: $q_1$ Early effect, $q_2$ high injection; ideal and non-ideal base diodes; $r_B$, $r_E$, $r_C$. |
| 31 | 17-18 4(a) | 8 | Ebers-Moll | Two diodes $I_F = I_{ES}(e^{V_{BE}/V_t} - 1)$, $I_R = I_{CS}(e^{V_{BC}/V_t} - 1)$ and sources $\alpha_F I_F$, $\alpha_R I_R$; $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$. Forward active: $I_C = \beta_F I_B$; saturation: both diodes on; cut-off; inverse. |
| 32 | 17-18 4(b) | 3 | Bandgap narrowing | At high doping the impurity levels form a band that merges with $E_c$; many-body effects lower the edges; $E_g = E_{g0} - \Delta E_g$ (about 80 meV at $10^{19}$ cm$^{-3}$ in Si); $p_{E0}$ rises by $e^{\Delta E_g/kT}$, so $\gamma$ and $\beta$ fall. |
| 33 | 17-18 5(a) | 4 | TEBT | Thin (2–5 nm) AlGaAs tunnel barrier between $n^+$ emitter and $p^+$ base. Electrons (small $m^*$, small $\Delta E_c$) tunnel through; holes (heavy, larger $\Delta E_v$) are reflected. $\gamma \to 1$ with a heavily doped base: high gain, low base resistance, low turn-on voltage. |
| 34 | 17-18 5(c) | 11 | HBT | Kirk effect: $n_c = J_C/(ev_{sat})$ cancels $N_C$; for $n_c > N_C$ the field peak moves to the $n^-/n^+$ interface and the base widens to $W_B + W_C$; $\beta$, $f_T$ fall; $J_K \approx ev_{sat}N_C$. HBT: heavy base doping lets $N_C$ rise (DHBT). Abrupt: spike, higher turn-on, gain $\exp(\Delta E_v/kT)$, hot-electron launch; graded: no spike, full $\exp(\Delta E_g/kT)$, lower turn-on, higher $\beta$. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| **JFETs** | | | | |
| 35 | 20-21 5(a) | 6 | Channel profiles | The gate–channel reverse bias grows from source to drain, so the depletion layer is a wedge, thickest at the drain. (i) $V_d < V_{d,sat}$: channel open throughout, linear region. (ii) $V_d = V_{d,sat}$: the depletion just touches the bottom at the drain (pinch-off). (iii) $V_d > V_{d,sat}$: the pinched-off part grows towards the source, current saturates. |
| 36 | 17-18 6(d) | 6 | Transconductance | $g_m = \partial I_D/\partial V_{GS}$ at constant $V_{DS}$; saturation $g_{ms} = -(2I_{DSS}/V_p)(1 - V_{GS}/V_p)$. Small-signal circuit: $C_{gs} \parallel r_{gs}$, $C_{gd} \parallel r_{gd}$, $g_m v'_{gs}$ with $r_{ds}$, $C_{ds}$, series $r_s$, $r_d$; with $r_s$: $g'_m = g_m/(1 + g_m r_s)$. |

# 6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Band structure and materials** | | | | |
| 1 | 20-21 1(c) | 6 | Hole bands | Parabolic band $E = E_v - \hbar^2k^2/2m^*$; slope $\hbar^2k/m^*$; ratio of slopes at the same $k$ is $m_{hh}/m_{lh}$; Luttinger masses along [100] ($\gamma_2$) and [111] ($\gamma_3$); ratios 1.45 and 5.30, both above 1. |
| 2 | 20-21 TT1 7(c) | 4 | Hole bands | Parabolic band $E = E_v - \hbar^2k^2/2m^*$; $\lvert dE/dk\rvert = \hbar^2k/m^*$; ratio at the same $k$ is $m_{hh}/m_{lh}$; Luttinger masses along [100] ($\gamma_2$) and [111] ($\gamma_3$); ratios 1.45 and 5.30, both above 1. |
| 3 | 17-18 1(b) | 4 | Effective mass | Band edge as a parabola (Taylor expansion); compare with $E = \hbar^2k^2/2m$ to get $1/m^* = (1/\hbar^2)d^2E/dk^2$; check with $a = (1/\hbar)(d^2E/dk^2)(dk/dt)$ and $\hbar\, dk/dt = F$, so $F = m^*a$; method: fit a parabola (finite difference) to E–k points at the extremum. |
| **Metal-semiconductor contacts** | | | | |
| 4 | 21-22 TT1 7(c) | 9 | Schottky junction | Assumptions (ideal contact, uniform $N_d$, depletion approximation, metal a perfect conductor); $\rho = +eN_d$ for $0 < x < x_n$, sheet $-eN_dx_n$ on the metal; Poisson $dE/dx = eN_d/\epsilon_s$, integrate, $E(x_n) = 0$: $E = -(eN_d/\epsilon_s)(x_n - x)$; $E = 0$ in the neutral region and in the metal (Gauss check at $x = 0$); $\lvert E_{max}\rvert = eN_dx_n/\epsilon_s$; $V_{bi} = eN_dx_n^2/2\epsilon_s$ gives $x_n$. |
| **Heterojunctions** | | | | |
| 5 | 20-21 1(d) | 8 | Band diagrams | $E_g$ of AlGaAs from $x$, $\Delta E_c = \Delta E_v = 0.156$ eV; band diagrams before (common vacuum level) and after contact (flat $E_F$, spike and step); hole concentration on each side; solve for $E_F$; $eV_{bi} = E_{FN} - E_{FP}$; put $E_{vN} - E_{vP} = -\Delta E_v$. |
| 6 | 20-21 4(a) | 12 | Fields and potentials | $\rho = +eN_{d1}$, $-eN_{a2}$; Poisson in each material with its own $\epsilon$; fields zero at the edges; $\epsilon_1E_1(0) = \epsilon_2E_2(0)$ gives $N_{d1}x_n = N_{a2}x_p$; integrate again with $\phi$ continuous; $V_{bi} = V_{bi1} + V_{bi2}$; solve for $x_n$, $x_p$. |
| 7 | 20-21 TT1 7(d) | 10 | Band diagrams | $E_g$ of AlGaAs from $x$, $\Delta E_c = \Delta E_v = 0.156$ eV; band diagrams before (common vacuum level) and after contact (flat $E_F$, spike and step); hole concentration on each side; solve for $E_F$; $eV_{bi} = E_{FN} - E_{FP}$; put $E_{vN} - E_{vP} = -\Delta E_v$. |
| 8 | 18-19 2(b) | 7 | Fields and potentials | Poisson on each side with its own $\epsilon$; $E = 0$ at $-x_n$ and $x_P$; $\epsilon_nE_n(0) = \epsilon_PE_P(0)$ gives $N_{dn}x_n = N_{aP}x_P$; $V_{bin} = eN_{dn}x_n^2/2\epsilon_n$, $V_{biP} = eN_{aP}x_P^2/2\epsilon_P$; add; eliminate one width; $x_n$, $x_P$, $W$. |
| **MOS capacitor** | | | | |
| 9 | 18-19 4(d) | 6 | Threshold voltage | Flat band: $Q'_m = -Q'_{ss}$, $\phi_s = 0$, so $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$; threshold: $Q'_{mT} = \lvert Q'_{SD}(\max)\rvert - Q'_{ss}$, $\phi_s = 2\phi_{fp}$; $V_T = V_{FB} + \lvert Q'_{SD}(\max)\rvert/C_{ox} + 2\phi_{fp}$. |

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Bipolar transistors** | | | | |
| 10 | 20-21 6(a) | 9 | Ebers-Moll | Two diodes $I_F$, $I_R$ with sources $\alpha_F I_F$, $\alpha_R I_R$; KCL gives $I_C$, $I_E$, $I_B$; saturation: drop the $-1$ terms, $V_{BE}$ and $V_{BC}$ as logarithms; solve for $I_F$, $I_R$ from $I_C$, $I_B$; $V_{CE} = V_{BE} - V_{BC}$; reciprocity $I_{CS}/I_{ES} = \alpha_F/\alpha_R$. |
| 11 | 18-19 2(c) | 8 | Base minority carriers | Diffusion equation in the base; cosh/sinh general solution; law of the junction: $\delta n_B(0) = n_{B0}(e^{eV_{BE}/kT} - 1)$, $\delta n_B(x_B) = -n_{B0}$; solve for the constants; sinh identity; narrow base gives a straight line. |
| 12 | 18-19 4(b) | 7 | HBT | Short-base $J_{nE} = eD_{nB}n_{iB}^2 e^{eV_{BE}/kT}/(N_{AB}W_B)$; short-emitter $J_{pE} = eD_{pE}n_{iE}^2 e^{eV_{BE}/kT}/(N_{DE}W_E)$; ratio; $n_{iB}^2/n_{iE}^2 = e^{\Delta E_g/kT}$. |
| 13 | 18-19 5(a) | 7 | Injection efficiency | Diffusion equation in the emitter with $\delta p_E(x_E) = 0$; $J_{pE}$ with $1/\tanh(x_E/L_E)$; base solution gives $J_{nE}$ with $1/\tanh(x_B/L_B)$; $\gamma = J_{nE}/(J_{nE} + J_{pE})$; drop small terms; short-region form. |
| 14 | 17-18 4(c) | 9 | Breakdown | $I_{CEO} = M(\alpha I_{CEO} + I_{CBO})$; $I_{CEO} = MI_{CBO}/(1 - \alpha M)$; breakdown at $\alpha M = 1$; $M = 1/[1 - (V/BV_{CBO})^n]$; $BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} = BV_{CBO}/\sqrt[n]{\beta}$. |
| **JFETs** | | | | |
| 15 | 20-21 5(b) | 8 | Pinch-off | Poisson with $\rho = +eN_d$ in the depletion layer; $E = 0$ at $y = h$; potential $eN_dh^2/2\epsilon_s$; equals $V_{bi} - V_{gs}$; gives $h$; pinch-off at $h = a$; $V_{p0} = ea^2N_d/2\epsilon_s$, $V_p = V_{bi} - V_{p0}$. |
| 16 | 18-19 6(d) | 4 | Cut-off frequency | $f_T = g_m/2\pi C_G$; $g_m(\max) = e\mu_n N_d W a/L$; $C_G(\min) = \epsilon_s WL/a$; $f_T(\max) = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) \propto N_d$. |
| 17 | 17-18 6(b) | 6 | Pinch-off | Poisson with $\rho = +eN_d$; $E(h) = 0$ gives $E = -eN_d(h-y)/\epsilon_s$; $\phi(h) = eN_dh^2/2\epsilon_s = V_{bi} - V_{gs}$; pinch-off at $h = a$: $V_{p0} = ea^2N_d/2\epsilon_s$. |

# 7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| **Alloys and strain** | | | | | |
| 1 | 20-21 2(d) | 7 | Quaternary alloys | Read the corner binaries from Fig. 1; Vegard's law (bilinear interpolation) for $a(x,y)$ and $E_g(x,y)$ of $In_xGa_{1-x}As_ySb_{1-y}$; set $a = 5.93$ Å and $E_g = 1$ eV; solve the two equations. | $x \approx 0.24$, $y \approx 0.60$ (straight-line interpolation; with the bowing printed in Fig. 1 no alloy has both values) |
| 2 | 17-18 1(d) | 9 | Quaternary alloys | $E_g = hc/\lambda$; Vegard bilinear $a(x,y)$ and $E_g(x,y)$ from the table; lattice condition $a = 5.93$ Å gives $y(x)$; $E_g$ along it; closest point; $f$ on InP, $h_c \approx a_s/(2f)$. | $E_g = 1.0$ eV not reachable on $a = 5.93$ Å; closest $x = 0$, $y = 0.36$ ($E_g = 0.965$ eV); $f = 1.04$ %, $h_c \approx 28$ nm |
| 3 | 17-18 TT1 7(a) | 8 | Lattice matching | $h_c = a_s/(2f)$ gives $f$; $a_L = a_s(1+f)$; Vegard $x = (a_L - a_{GaAs})/(a_{InAs} - a_{GaAs})$. | $f = 5.65$ %, $a_L = 5.973$ Å, $x = 0.79$ |
| **Metal-semiconductor contacts** | | | | | |
| 4 | 20-21 3(c) | 8 | Schottky junction | $\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_dx_n/\epsilon_s$; $C' = \epsilon_s/x_n$. | $\phi_{B0} = 0.54$ V; $V_{bi} = 0.334$ V; $\lvert E_{max}\rvert = 3.22 \times 10^4$ V/cm; $C' = 4.98 \times 10^{-8}$ F/cm$^2$ |


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| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| 5 | 20-21 3(d) | 6 | Barrier lowering | Image-force lowering $\Delta\phi = \sqrt{eE/(4\pi\epsilon_s)}$; position $x_m = \sqrt{e/(16\pi\epsilon_s E)}$ with $\epsilon_s$ of GaAs. | $\Delta\phi = 27.3$ mV at $x_m = 2.01$ nm |
| 6 | 18-19 1(c) | 6 | Schottky junction | $\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$. | $\phi_{B0} = 0.97$ V; $V_{bi} = 0.790$ V; $\lvert E_{max}\rvert = 4.23 \times 10^4$ V/cm |
| 7 | 18-19 3(d) | 0 | Schottky junction | n-type: $\phi_{Bn} = \phi_m - \chi$; p-type: $\phi_{Bp} = E_g/e + \chi - \phi_m$; positive barrier means Schottky, zero or negative means ohmic. | Al/n-GaAs ohmic ($-0.01$ V); Cu/n-GaAs Schottky 0.91 V; Ag/p-GaAs Schottky 0.849 V; Au/p-GaAs Schottky 0.119 V (nearly ohmic) |
| 8 | 18-19 TT1 7(b) | 6 | Schottky junction | $\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$. | $\phi_{B0} = 0.97$ V; $V_{bi} = 0.790$ V; $\lvert E_{max}\rvert = 4.23 \times 10^4$ V/cm |
| 9 | 17-18 2(c) | 10 | Schottky junction | $\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$. | $\phi_{B0} = 0.48$ V; $V_{bi} = 0.362$ V; $x_n = 0.324$ $\mu$m; $\lvert E_{max}\rvert = 2.24 \times 10^4$ V/cm |
| **Heterojunctions** | | | | | |
| 10 | 18-19 2(d) | 5 | Fields and potentials | $\Delta E_c = \chi_n - \chi_P$, $\Delta E_v = \Delta E_g - \Delta E_c$; solve $eV_{bi} = \Delta E_v + kT \ln(N_{aP}N_{vn}/(p_{n0}N_{vP}))$ for $p_{n0}$; $N_d = n_i^2/p_{n0}$; $x_P$ from the heterojunction formula. | $\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV; $N_d = 7.2 \times 10^{15}$ cm$^{-3}$; $x_P = 0.261$ $\mu$m |
| 11 | 17-18 2(a) | 7 | Band diagrams | Anderson: $\Delta E_c = \chi_n - \chi_P$, $\Delta E_v = \Delta E_g - \Delta E_c$; $p_{n0} = n_i^2/N_d$; $eV_{bi} = \Delta E_v + kT \ln(p_{P0}N_{vn}/(p_{n0}N_{vP}))$. | $\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV, $V_{bi} = 1.01$ V |
| 12 | 17-18 5(d) | 5 | HEMT | $V_{p2} = eN_d d_d^2/(2\epsilon_N)$; $V_{off} = \phi_B - \Delta E_c/e - V_{p2}$; $n_s = \epsilon_N(V_g - V_{off})/[e(d_d + d_i + \Delta d)]$ with $\Delta d = 80$ Å. | $V_{p2} = 1.85$ V; $V_{off} = -1.22$ V; $n_s = 1.37 \times 10^{12}$ cm$^{-2}$ |
| **MOS capacitor** | | | | | |
| 13 | 18-19 TT2 8(b) | 5 | Threshold voltage | $\phi_{fp} = V_t \ln(N_a/n_i)$; $x_{dT} = (4\epsilon_s\phi_{fp}/eN_a)^{1/2}$; $\lvert Q'_{SD}(\max)\rvert = eN_a x_{dT}$; $V_{TN} = (\lvert Q'_{SD}(\max)\rvert - eQ'_{ss})t_{ox}/\epsilon_{ox} + \phi_{ms} + 2\phi_{fp}$. | $\phi_{fp} = 0.228$ V, $x_{dT} = 2.43$ $\mu$m, $V_{TN} = 0.546$ V |
| **Bipolar transistors** | | | | | |
| 14 | 18-19 5(d) | 5 | Injection efficiency | $L = \sqrt{D\tau}$; $\alpha = \beta/(1+\beta)$; take the minimum $x_B$, $\alpha_T = 1/\cosh(x_B/L_B)$; $\gamma = \alpha/(\alpha_T\delta)$; solve the $\gamma$ formula for $N_E$. | $x_B = 0.80$ $\mu$m, $N_E = 4.6 \times 10^{18}$ cm$^{-3}$; $\alpha_T = 0.99930$, $\gamma = 0.99656$ |
| 15 | 17-18 4(d) | 5 | Bandgap narrowing | $p_{E0} = n_i^2/N_E$; read $\Delta E_g$ from Fig. 1; $p'_{E0} = p_{E0}\exp(\Delta E_g/kT)$. | $\Delta E_g \approx 23$ and 78 meV; $p'_{E0} = 549$ and 454 cm$^{-3}$; ratio 2.4 and 20 |
| **JFETs** | | | | | |
| 16 | 20-21 5(c) | 7 | Pinch-off | $V_{bi} = V_t \ln(N_a N_d/n_i^2)$; $V_{p0} = ea^2 N_d/2\epsilon_s$; $V_p = V_{bi} - V_{p0}$; for an open channel $a - h = 0.10$ $\mu$m, $h = 0.60$ $\mu$m, $V_{bi} - V_{gs} = eN_d h^2/2\epsilon_s$, solve for $V_{gs}$. | $V_{bi} = 1.250$ V; $V_{p0} = 1.016$ V; $V_p = +0.235$ V; $V_{gs} \approx +0.50$ V (forward) |


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| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| 17 | 20-21 6(c) | 8 | Transconductance | $I_{P1}=\mu_n(eN_d)^2Wa^3/(6\epsilon_sL)$; $V_{bi}$ and $V_{p0}$; $I_D(sat)$ at $V_{GS}=0$ from the JFET saturation equation; $g_{ms}=(3I_{P1}/V_{p0})(1-\sqrt{(V_{bi}-V_{GS})/V_{p0}})$ at $V_{GS}=0$. | $I_{P1}=0.522$ mA ($I_{D1}(sat)=0.3135$ mA at $V_{GS}=0$); $g_{ms}(max)=0.204$ mS |
| 18 | 18-19 6(e) | 5 | Pinch-off | $\phi_n=V_t\ln(N_c/N_d)$; $V_{bi}=\phi_{Bn}-\phi_n$; $V_{p0}=V_{bi}-V_T$; $a=(2\epsilon_sV_{p0}/eN_d)^{1/2}$. | $V_{bi}=0.749$ V, $V_{p0}=0.649$ V, $a=0.686$ $\mu$m |
| 19 | 17-18 6(c) | 5 | Pinch-off | $V_{p0}=ea^2N_a/(2\epsilon_s)$; $V_{bi}=V_t\ln(N_aN_d/n_i^2)$; p-channel: $V_p=V_{p0}-V_{bi}$. | $V_{p0}=1.236$ V; $V_{bi}=0.814$ V; $V_p=0.422$ V |

# 8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

| No. | Paper | Kind | Diagram |
|---|---|---|---|
| **Band structure and materials** | | | |
| 1 | 21-22 TT1 7(b) | Diagram | bulk, quantum well, wire and dot |
| 2 | 21-22 TT1 7(b) | Graph | density of states against energy in 3D, 2D, 1D and 0D |
| 3 | 20-21 1(c) | Graph | E-k of the Si heavy- and light-hole bands along [111] and [100] |
| 4 | 20-21 2(a) | Graph | fibre loss windows and the wavelength ranges of the materials |
| 5 | 20-21 3(b) | Diagram | counting allowed $k$ points in 1D, 2D and 3D |
| 6 | 20-21 3(b) | Graph | density of states against energy in 3D, 2D, 1D and 0D |
| 7 | 20-21 TT1 7(c) | Graph | E-k of the Si heavy- and light-hole bands along [111] and [100] |
| 8 | 18-19 1(a) | Diagram | hexagonal close-packed cell |
| 9 | 18-19 1(a) | Diagram | zinc blende (GaAs) cell with tetrahedral bonds |
| 10 | 18-19 1(a) | Diagram | body-centred cubic cell |
| 11 | 17-18 1(a) | Diagram | atoms of a single-crystal, a polycrystalline and an amorphous solid |
| 12 | 17-18 1(b) | Graph | two E–k parabolas of different curvature |
| 13 | 17-18 1(b) | Graph | parabola fitted to E–k points |
| **High-field transport** | | | |
| 14 | 20-21 1(a) | Diagram | electron drift with and without a field |
| 15 | 20-21 1(a) | Diagram | simplified E-k diagrams of Si and GaAs |
| 16 | 20-21 1(b) | Graph | drift velocity against field for Si and GaAs |
| 17 | 20-21 1(b) | Diagram | the two GaAs valleys and electron transfer |
| 18 | 20-21 1(b) | Diagram | a travelling domain in a Gunn diode and the current pulses |
| 19 | 20-21 TT1 7(a) | Diagram | simplified E-k diagrams of Si and GaAs |
| 20 | 20-21 TT1 7(b) | Diagram | the two GaAs valleys and electron transfer |
| 21 | 20-21 TT1 7(b) | Diagram | a travelling domain in a Gunn diode and the current pulses |
| 22 | 17-18 TT1 7(b) | Graph | E–k of Si and GaAs |
| 23 | 17-18 TT1 7(b) | Graph | drift velocity against field for Si and GaAs |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| **Alloys and strain** | | | |
| 24 | 21-22 TT1 7(a) | Graph | direct and indirect gaps of GaAsP against x |
| 25 | 21-22 TT1 7(a) | Diagram | simplified E-k of GaAs, the crossover and GaP |
| 26 | 20-21 2(b) | Diagram | matched, strained and relaxed layers on a substrate |
| 27 | 20-21 2(b) | Graph | critical thickness against misfit |
| 28 | 20-21 2(c) | Graph | GaAsP band gap against $x$ with direct and indirect branches |
| 29 | 20-21 2(d) | Graph | InGaAsSb region of Fig. 1 with the target point |
| 30 | 18-19 1(d) | Diagram | horizontal VPE reactor with its temperature profile |
| 31 | 17-18 1(c) | Graph | $E_\Gamma$ and $E_X$ of $GaAs_{1-x}P_x$ against $x$ |
| 32 | 17-18 1(d) | Graph | lattice and band-gap lines in the $(x, y)$ square |
| 33 | 17-18 3(b) | Diagram | matched, strained and relaxed layers |
| 34 | 17-18 3(b) | Diagram | E–k of Si and GaAs with the transitions |
| **PN junction** | | | |
| 35 | 20-21 4(c) | Diagram | graded junction by diffusion |
| 36 | 20-21 4(c) | Graph | diffused acceptor profile and net doping |
| 37 | 20-21 4(c) | Graph | linear charge, parabolic field |
| **Metal-semiconductor contacts** | | | |
| 38 | 21-22 TT1 7(c) | Diagram | metal and n-semiconductor before contact and in equilibrium |
| 39 | 21-22 TT1 7(c) | Graph | charge, field and potential across the junction |
| 40 | 20-21 3(a) | Diagram | metal–n-semiconductor band diagrams at zero, reverse and forward bias |
| 41 | 20-21 3(c) | Diagram | W and n-Si before and after contact |
| 42 | 20-21 3(c) | Graph | space charge and field in the depletion region |
| 43 | 20-21 3(d) | Diagram | electron and its image charge, and the lowered barrier |
| 44 | 20-21 4(b) | Graph | barrier with image-force lowering |
| 45 | 20-21 4(b) | Diagram | metal–semiconductor with an interfacial layer and interface states |
| 46 | 20-21 4(b) | Graph | barrier height against metal work function |
| 47 | 20-21 5(d) | Diagram | metal and n-semiconductor band diagrams for $\phi_m < \phi_s$ |
| 48 | 20-21 5(d) | Graph | ohmic and rectifying I-V curves |
| 49 | 18-19 1(b) | Diagram | metal–semiconductor band diagram with an interfacial layer and interface states |
| 50 | 18-19 1(c) | Diagram | Au / n-Ge band diagram with the values |
| 51 | 18-19 1(c) | Graph | electric field in the depletion region |
| 52 | 18-19 3(a) | Graph | specific contact resistance against doping for several barriers |
| 53 | 18-19 3(c) | Diagram | ohmic contacts by accumulation and by tunnelling |
| 54 | 18-19 3(d) | Diagram | band diagrams of the four metal–GaAs junctions |
| 55 | 18-19 TT1 7(a) | Diagram | ohmic contacts by accumulation and by tunnelling |
| 56 | 18-19 TT1 7(b) | Diagram | Au / n-Ge band diagram with the values |
| 57 | 17-18 2(c) | Diagram | W / n-GaAs band diagram with the values |
| 58 | 17-18 3(a) | Diagram | metal and n-type semiconductor before and after contact ($\phi_m > \phi_s$) |
| 59 | 17-18 3(a) | Diagram | metal and p-type semiconductor before and after contact ($\phi_m < \phi_s$) |
| 60 | 17-18 3(c) | Diagram | tunnelling junction band diagram |
| **Heterojunctions** | | | |
| 61 | 20-21 1(d) | Diagram | the three heterojunction band alignments |
| 62 | 20-21 1(d) | Diagram | p-GaAs and N-AlGaAs before contact |
| 63 | 20-21 1(d) | Diagram | the heterojunction after contact in equilibrium |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 64 | 20-21 4(a) | Diagram | narrow-gap N and wide-gap P before contact |
| 65 | 20-21 4(a) | Diagram | equilibrium band diagram of the straddling nP junction |
| 66 | 20-21 4(a) | Graph | charge, field and potential with the field jump at $x = 0$ |
| 67 | 20-21 TT1 7(d) | Diagram | p-GaAs and N-AlGaAs before contact |
| 68 | 20-21 TT1 7(d) | Diagram | the heterojunction after contact in equilibrium |
| 69 | 18-19 2(a) | Diagram | pN (p-GaAs/N-AlGaAs) equilibrium band diagram |
| 70 | 18-19 2(a) | Diagram | nN (n-GaAs/N-AlGaAs) band diagram with a notch |
| 71 | 18-19 2(b) | Graph | charge, field and potential of a heterojunction |
| 72 | 18-19 2(d) | Diagram | n-Ge / P-GaAs band diagram with the values |
| 73 | 18-19 3(b) | Diagram | RTD band diagrams at four biases |
| 74 | 18-19 3(b) | Graph | I-V with two peaks |
| 75 | 18-19 6(b) | Diagram | HEMT layer structure |
| 76 | 18-19 6(b) | Diagram | conduction band under the gate with the 2DEG |
| 77 | 17-18 2(a) | Diagram | n-Ge / p-GaAs equilibrium band diagram |
| 78 | 17-18 2(b) | Diagram | $N^+$-AlGaAs / intrinsic GaAs band diagram with a 2DEG notch |
| 79 | 17-18 2(b) | Diagram | $P^+$-AlGaAs / n-GaAs band diagram |
| 80 | 17-18 5(b) | Diagram | HEMT layer structure |
| 81 | 17-18 5(b) | Diagram | conduction band under the gate with the 2DEG well |
| **MOS capacitor** | | | |
| 82 | 18-19 4(c) | Graph | charge distribution in an MOS capacitor at inversion |
| 83 | 18-19 4(c) | Graph | electric field and potential across oxide and silicon |
| 84 | 18-19 4(c) | Graph | high- and low-frequency C-V curves |
| 85 | 18-19 4(d) | Diagram | MOS band diagrams at flat band and at threshold |
| 86 | 18-19 TT2 8(a) | Diagram | MOS band diagrams in accumulation, depletion and inversion for p- and n-type substrates |
| **Bipolar transistors** | | | |
| 87 | 20-21 6(a) | Diagram | npn regions and currents |
| 88 | 20-21 6(a) | Circuit | Ebers-Moll equivalent circuit |
| 89 | 20-21 6(a) | Graph | output curves with the saturation region, and $V_{CE}(sat)$ against $I_B$ |
| 90 | 20-21 6(b) | Graph | Gummel plot with high-level injection |
| 91 | 20-21 6(b) | Diagram | emitter current crowding under the emitter |
| 92 | 20-21 6(b) | Diagram | interdigitated emitter-base layout |
| 93 | 18-19 2(c) | Graph | minority carriers in E, B, C in forward-active mode |
| 94 | 18-19 2(c) | Graph | exact base profile for several $x_B/L_B$ |
| 95 | 18-19 4(a) | Diagram | DHBT mesa layer structure |
| 96 | 18-19 4(a) | Graph | DHBT doping profile and Al fraction |
| 97 | 18-19 4(b) | Diagram | HBT band diagram in forward-active mode |
| 98 | 18-19 5(a) | Diagram | current components in an npn transistor |
| 99 | 18-19 5(b) | Graph | base-width modulation and output curves with the Early voltage |
| 100 | 18-19 5(b) | Diagram | emitter current crowding |
| 101 | 18-19 5(c) | Circuit | Gummel-Poon equivalent circuit |
| 102 | 18-19 5(c) | Graph | Gummel plot and gain against current |
| 103 | 18-19 6(c) | Circuit | Ebers-Moll equivalent circuit |
| 104 | 17-18 4(a) | Circuit | Ebers–Moll equivalent circuit |
| 105 | 17-18 4(a) | Diagram | npn modes of operation |
| 106 | 17-18 4(c) | Graph | $I_C$ against $V_{CE}$ showing $BV_{CEO} < BV_{CBO}$ |
| 107 | 17-18 4(d) | Graph | bandgap narrowing against donor doping (Fig. 1) |
| 108 | 17-18 5(a) | Diagram | TEBT band diagram in forward-active mode |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 109 | 17-18 5(c) | Graph | charge and field in the collector at low and high current |
| 110 | 17-18 5(c) | Diagram | bands with base push-out |
| 111 | 17-18 5(c) | Diagram | abrupt and graded HBT emitter-base junctions |
| **JFETs** | | | |
| 112 | 20-21 5(a) | Diagram | channel space-charge profiles in the three regions |
| 113 | 20-21 5(a) | Graph | drain current against drain voltage with the three points |
| 114 | 20-21 5(b) | Diagram | channel geometry of single- and two-gate JFETs |
| 115 | 20-21 5(b) | Graph | charge, field and potential across the gate junction |
| 116 | 20-21 5(c) | Graph | depletion width against gate voltage |
| 117 | 20-21 6(c) | Graph | transfer curve and transconductance |
| 118 | 18-19 6(d) | Diagram | MESFET cross-section with $a$, $L$, $h$ |
| 119 | 17-18 6(a) | Diagram | JFET depletion region at $V_d = 0$, below, at and above $V_{d,sat}$ |
| 120 | 17-18 6(b) | Diagram | MESFET cross-section with $a$ and $h$ |
| 121 | 17-18 6(d) | Circuit | JFET small-signal equivalent circuit |

# 9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

| Pair | Key difference | Asked in |
|---|---|---|
| **Band structure and materials** | | |
| Single crystal vs polycrystalline vs amorphous | long-range order everywhere vs within grains vs none | 17-18 1(a) |
| hcp vs bcc | close packed, 12 neighbours, packing 0.74 vs 8 neighbours, packing 0.68 | 18-19 1(a) |
| 3D vs 2D vs 1D density of states | $\sqrt{E}$ vs steps vs $1/\sqrt{E}$ spikes | 21-22 TT1 7(b), 20-21 3(b) |
| **High-field transport** | | |
| Si vs GaAs electrons | indirect gap, six flat X valleys, $m^* = 0.26\,m_0$ vs direct gap, one curved $\Gamma$ valley, $m^* = 0.067\,m_0$, 6 times the mobility | 20-21 1(a), 20-21 TT1 7(a), 17-18 TT1 7(b) |
| Velocity saturation vs Gunn effect | velocity levels off (Si) vs velocity falls after a peak, negative mobility, oscillation (GaAs) | 20-21 1(b) |
| **Alloys and strain** | | |
| Direct vs indirect gap | minimum at $k = 0$, vertical transition, bright vs minimum at the zone edge, needs a phonon, dim | 21-22 TT1 7(a), 17-18 3(b) |
| Strained vs relaxed layer | thinner than $h_c$, no dislocations, distorted cell vs thicker, misfit dislocations, own lattice constant | 20-21 2(b) |
| **PN junction** | | |
| Abrupt vs linearly graded junction | step doping, $W \propto V^{1/2}$ vs linear doping, $W \propto V^{1/3}$ | 20-21 4(c) |
| **Metal-semiconductor contacts** | | |
| Rectifying on n vs p | $\phi_m > \phi_s$, bands bend up vs $\phi_m < \phi_s$, bands bend down | 17-18 3(a) |
| Schottky limit vs Bardeen limit | barrier follows $\phi_m - \chi$ vs barrier pinned by interface states | 20-21 4(b), 18-19 1(b) |
| Barrier-free vs tunnelling ohmic contact | no barrier vs thin barrier from heavy doping | 18-19 TT1 7(a) |
| Reverse vs forward biased Schottky junction | barrier $V_{bi} + V_R$, wider depletion, small current vs barrier $V_{bi} - V_a$, narrower, large electron current | 20-21 3(a) |


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| Pair | Key difference | Asked in |
|---|---|---|
| Ohmic vs rectifying contact | $\phi_m < \phi_s$ (n-type), no barrier, linear I-V vs $\phi_m > \phi_s$, Schottky barrier, diode I-V | 20-21 5(d) |
| **Heterojunctions** | | |
| $N^+$/i vs $P^+$/n AlGaAs/GaAs | accumulation notch in the GaAs vs depletion of the n-GaAs | 17-18 2(b) |
| pN vs nN heterojunction | both sides depleted vs depletion plus accumulation (2DEG notch) | 18-19 2(a) |
| HEMT vs MESFET | 2DEG in undoped GaAs vs electrons among their donors | 18-19 6(b) |
| Straddling vs staggered vs broken-gap heterojunction | one gap inside the other vs offsets in the same direction vs the gaps do not overlap | 20-21 1(d) |
| **MOS capacitor** | | |
| High- vs low-frequency C-V in inversion | stays at $C'_{min}$ vs returns to $C_{ox}$ | 18-19 4(c) |
| **Bipolar transistors** | | |
| Abrupt vs graded HBT | spike, higher turn-on, hot electrons vs no spike, lower turn-on, higher gain | 17-18 5(c) |
| SHBT vs DHBT | wide-gap emitter only vs wide-gap emitter and collector | 18-19 4(a) |
| BJT vs HBT | gain needs $N_E \gg N_B$ vs wide-gap emitter, heavily doped base | 18-19 4(b) |
| Early effect vs current crowding | base width changes with $V_{CB}$ vs lateral IR drop under the emitter | 18-19 5(b) |
| Ebers-Moll vs Gummel-Poon | constant two-diode model vs charge-based with Early effect and high injection | 18-19 5(c) |
| High-level injection vs current crowding | base charge rises, $\beta$ falls vs lateral IR drop, current concentrates at emitter edges | 20-21 6(b) |

# 10. Repeated questions:

12 questions came back, exactly or with new numbers; 1 of them in three or more papers. Most repeated first; the latest paper first in each row.

| No. | Times | Topic | Subtopic | Asked in (marks) |
|---|---|---|---|---|
| 1 | 3 | High-field transport | Mobility | 20-21 1(a) (6), 20-21 TT1 7(a) (3), 17-18 TT1 7(b) (12) |
| 2 | 2 | Alloys and strain | GaAsP band gaps | 21-22 TT1 7(a) (3), 17-18 1(c) (9) |
| 3 | 2 | Band structure and materials | Density of states | 21-22 TT1 7(b) (8), 20-21 3(b) (6) |
| 4 | 2 | JFETs | Channel profiles | 20-21 5(a) (6), 17-18 6(a) (8) |
| 5 | 2 | JFETs | Pinch-off | 20-21 5(b) (8), 17-18 6(b) (6) |
| 6 | 2 | High-field transport | Gunn effect | 20-21 1(b) (5), 20-21 TT1 7(b) (3) |
| 7 | 2 | Band structure and materials | Hole bands | 20-21 1(c) (6), 20-21 TT1 7(c) (4) |
| 8 | 2 | Heterojunctions | Band diagrams | 20-21 1(d) (8), 20-21 TT1 7(d) (10) |
| 9 | 2 | Bipolar transistors | Ebers-Moll | 18-19 6(c) (5), 17-18 4(a) (8) |
| 10 | 2 | Heterojunctions | HEMT | 18-19 6(b) (7), 17-18 5(b) (5) |
| 11 | 2 | Metal-semiconductor contacts | Schottky junction | 18-19 1(c) (6), 18-19 TT1 7(b) (6) |
| 12 | 2 | Metal-semiconductor contacts | Ohmic contacts | 18-19 3(c) (3), 18-19 TT1 7(a) (4) |


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# 11. Question style:

How each topic was asked, as a number of questions.

| Topic | Theory | Derivation | Numerical | Diagram |
|---|---|---|---|---|
| Band structure and materials | 4 | 3 | · | 1 |
| High-field transport | 5 | · | · | · |
| Alloys and strain | 6 | · | 3 | · |
| PN junction | 1 | · | · | · |
| Metal-semiconductor contacts | 7 | 1 | 6 | 2 |
| Heterojunctions | 4 | 4 | 3 | 2 |
| MOS capacitor | · | 1 | 1 | 2 |
| Bipolar transistors | 7 | 5 | 2 | 2 |
| JFETs | 2 | 3 | 4 | 1 |

# 12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

| No. | Topic | Marks | Covered so far |
|---|---|---|---|
| 1 | Bipolar transistors | 108 | 21 % |
| 2 | Metal-semiconductor contacts | 94 | 40 % |
| 3 | Heterojunctions | 81 | 55 % |
| 4 | Alloys and strain | 66 | 68 % |
| 5 | JFETs | 63 | 81 % |
| 6 | Band structure and materials | 42 | 89 % |
| 7 | High-field transport | 29 | 95 % |
| 8 | MOS capacitor | 22 | 99 % |
| 9 | PN junction | 5 | 100 % |


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